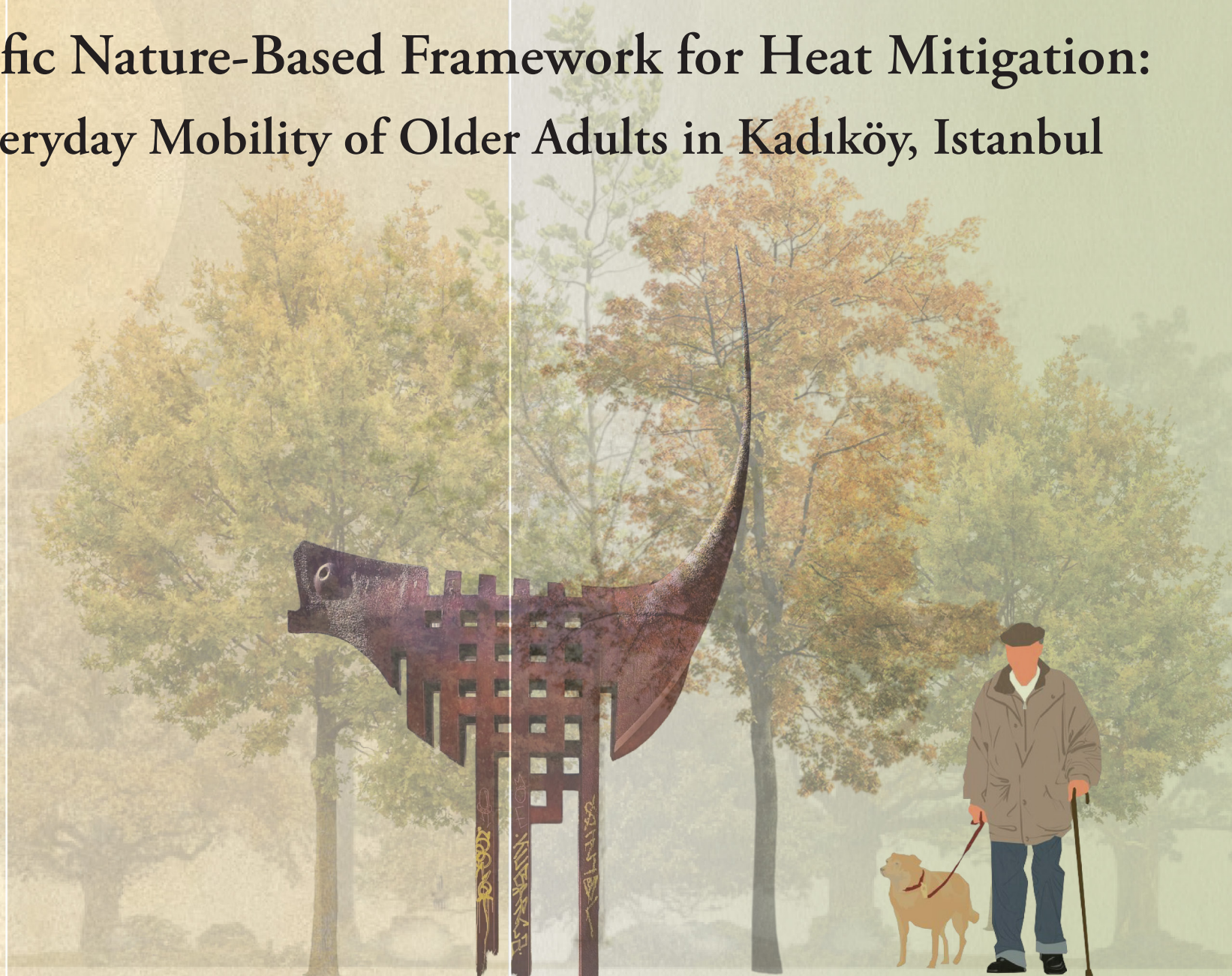
A Context-Specific Nature-Based Framework for Heat Mitigation:

Supporting the Everyday Mobility of Older Adults in Kadıköy, Istanbul



SUDes
Sustainable Urban Design

A Context-Specific Nature-Based Framework for Heat Mitigation

Supporting the Everyday Mobility of Older Adults in Kadıköy, Istanbul

Faculty of Engineering, LTH

ASBM01: Degree Project in Sustainable Urban Design

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Abstract

Climate change is intensifying summer heatwaves globally, and especially across the Mediterranean region, which is warming approximately 20% faster than the global average, while in cities their impact is further amplified by the Urban Heat Island effect. Heat affects different groups in different ways, depending on their exposure, their sensitivity, and their ability to respond and adapt; older adults are particularly affected, as they are physiologically more vulnerable to heat while also depending on walking to reach their everyday destinations. Although Nature-Based Solutions (NBS) are widely promoted to mitigate urban heat, their application is often generalised in practice: Istanbul's Climate Change Action Plan, for example, largely frames them as tree planting, with limited consideration of whether streets can accommodate trees or whom the cooling should serve. This thesis therefore asks how NBS can be contextualised to mitigate urban heat while supporting the everyday mobility of an ageing population.

Kadıköy, the Istanbul district with the largest older population, where 20.9% of residents are aged 65 or over, provides the case study. To identify where heat accumulates across the district, Land Surface Temperature derived from Landsat imagery was analysed alongside demographic data, revealing a heterogeneous distribution of heat across institutional campuses, sports facilities, transport infrastructure, commercial clusters detached from residential areas, and dense residential and mixed-use fabric. Since the study focuses on the everyday mobility of older adults, the analysis concentrated on the fabric where they live and walk, excluding neighbourhoods with the smallest proportion of older residents as well as those whose residents have a higher capacity to adapt to heat. On this basis, Caferağa and Osmanağa were selected, as their residential streets lie within walking distance of the district's most frequently visited destinations, while their dense, historic, and organically grown fabric constrains the space available for generic tree planting

As greening every street is neither logical nor possible, the everyday destinations of older adults, informed by an exploratory interview, were mapped together with the streets that connect them in order to identify where cooling should be prioritised. Their concentration along Bahariye Caddesi, Moda Caddesi, and Söğütlüçeşme Caddesi forms the basis of the Green Loop, which passes within 100 m of around two-thirds of the mapped destinations and is developed through four actions (Anchor, Cool, Connect, and Expand) that first concentrate cooling in the mixed-use commercial core and then extend it into the surrounding residential streets. Ten street sections along the loop were assessed by enclosure, orientation, street function, planting capacity, and existing vegetation, and grouped into four typologies, within each of which the section most exposed to direct sunlight was selected as a design site. Nature-Based Solutions were selected from the World Bank's catalogue according to the processes through which they cool, the scale at which they operate, and their suitability for dense urban fabric, and translated into a toolbox of seven street-scale tools. At each design site, these tools were matched to local conditions: where trees cannot be accommodated, climbing plants and vegetated pergolas are designed to provide shade, while permeable surfaces and the reallocation of parking and street space reduce exposed impervious surfaces.

The result is a proposed network of cooler everyday spaces and walking routes connecting the destinations on which older adults depend. The thesis thereby contributes a transferable framework in which the social context informs where to intervene and the physical context informs how. As the cooling performance of the designs was not simulated, future research should validate the interventions through microclimate modelling and involve older adults directly in the design process.

Keywords: urban heat island, Nature-Based Solutions, older adults, everyday mobility, outdoor thermal comfort, context-specific design, Istanbul

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I would like to dedicate this thesis to my grandfather, whom I recently lost. I would not have found my way to Lund without his support. Thank you, Baba.

I would also like to dedicate this work to my mother and my grandmother, who have always supported me throughout these two years.

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1. Introduction

1.1 Background

Urban heat is an increasing challenge for climate adaptation, particularly in the Mediterranean region, which is warming approximately 20% faster than the global average (MedECC, 2020; Figure 1.1). Heatwaves in the region are becoming more frequent, longer, and more intense (IPCC, 2022), and in cities their impact is amplified by the Urban Heat Island (UHI) effect (Li & Bou-Zeid, 2013). The UHI is driven by dense development, impervious surfaces, and limited vegetation, which absorb and retain solar radiation, as well as by anthropogenic heat from traffic, air conditioning, and other human activities (Oke, 1982; Sailor, 2011). As these drivers vary spatially, heat is unevenly distributed across the urban fabric.

The impacts of urban heat are also socially unequal, as heat-related risk depends on exposure, sensitivity, and the capacity to adapt (IPCC, 2022). Older adults are among the groups most vulnerable to extreme heat, as age-related changes in thermoregulation and a higher prevalence of chronic conditions increase their susceptibility to heat-related illness and mortality (WHO, 2025). At the same time, the mobility of older adults is closely tied to their immediate built environment, with many relying on walking to reach nearby destinations (Rosso et al., 2011). Adverse outdoor thermal conditions can therefore affect not only their health, but also their access to everyday services and their participation in public life. Urban heat adaptation thus requires consideration of both where heat accumulates and where cooling can support the everyday mobility of heat-vulnerable populations.

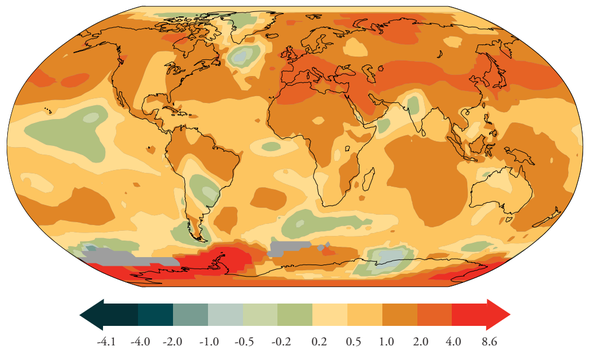


Figure 1.1. Global surface air temperature anomalies, June–August 2025, relative to the 1951–1980 baseline. Source: NASA GISS (2026).

1.2 Research Gap

Nature-Based Solutions (NBS) are increasingly recognised as an important approach to urban climate adaptation, including the mitigation of urban heat (Kabisch et al., 2016). Their cooling effect, however, depends on mechanisms such as shading and evapotranspiration, and varies with the type, size, and setting of green interventions (Bowler et al., 2010). Despite this, their application is often discussed through broadly applicable interventions rather than strategies tailored to the contexts in which they are implemented. Reviewing frameworks for NBS, Wickenberg et al. (2021) observe that “few frameworks for NBS address implementation”, and argue that NBS need to be “adapted to site-specific societal challenges”.

This tendency is also visible in local climate policy. In the Istanbul Climate Change Action Plan, NBS for heat mitigation are largely framed as tree planting (İBB, 2021, p. 58), with limited consideration of whether dense street environments can accommodate trees, or of whom the cooling should serve.

The suitability of cooling measures depends not only on physical conditions, but also on where cooling is most needed and who is expected to benefit from it. Frameworks for prioritising urban greening have begun to combine heat exposure with population vulnerability (Norton et al., 2015). This thesis builds on such approaches by focusing on the everyday destinations and walking routes of older adults, and on how NBS can be adapted to the physical constraints of these spaces. This creates a need for a context-specific approach to NBS that considers both the physical characteristics of the urban environment and the everyday spatial needs of heat-vulnerable populations.

1.3 Research Aim

This thesis aims to develop a context-specific framework for integrating Nature-Based Solutions into existing urban fabric to mitigate urban heat, responding both to the physical conditions of the urban environment and to the everyday spatial needs of heat-vulnerable populations. Using older adults as a social prioritisation lens, the framework investigates where urban cooling should be prioritised by considering the spatial distribution of heat alongside everyday destinations and walking routes, and how different NBS can be selected, adapted, and combined according to local conditions.

The intention is not to develop a universally applicable catalogue of cooling interventions, but an approach in which the social context informs where to intervene and the physical context informs how.

1.4 Research Objectives

1. To identify the spatial distribution of urban heat within a dense urban district and determine where it coincides with the urban fabric in which older adults live and walk.

2. To identify the everyday destinations and walking routes of older adults and establish where urban cooling interventions should be prioritised.

3. To examine local spatial and environmental conditions, including street morphology, orientation, solar exposure, vegetation, and available space, in order to identify where intervention is needed and to select and adapt appropriate NBS.

4. To translate these findings into a connected and phased urban cooling strategy that integrates context-specific NBS along everyday destinations and walking routes.

5. To reflect on the transferability of the approach to other urban contexts and other heat-vulnerable populations.

1.5 Research Question

How can Nature-Based Solutions be contextualised to mitigate urban heat while supporting the everyday mobility of an ageing population?

Guiding Questions

1. Where should urban cooling be prioritised when patterns of urban heat are considered alongside the everyday destinations and walking routes of older adults?

2. How do local spatial and environmental conditions influence the selection and adaptation of Nature-Based Solutions?

3. How can context-specific Nature-Based Solutions be integrated into a connected cooling network that supports everyday mobility?

1.6 Thesis Structure

The thesis is organised into nine chapters, moving from the general to the specific and back again. Following this introduction, Chapter 2 reviews the literature on urban heat in the Mediterranean, the Urban Heat Island effect and its drivers, heat vulnerability and the outdoor thermal comfort of older adults, and the cooling performance of Nature-Based Solutions (NBS) in different urban settings. Chapter 3 sets out the research design and the five stages of the methodology, from district-scale heat analysis to site-scale design.

Chapter 4 narrows the focus from Istanbul to the district of Kadıköy, analysing its population, climate, and distribution of surface heat, explaining the selection of Caferağa and Osmanağa, and examining their urban conditions and the everyday destinations of older adults. Building on this analysis, Chapter 5 develops the Green Loop, a phased strategy connecting everyday destinations, and classifies ten street sections along it into four typologies, selecting the most sun-exposed section of each as a design site. Chapter 6 presents the NBS toolbox and explains how its seven tools were derived from the World Bank's catalogue of Nature-Based Solutions, and Chapter 7 applies it to the design sites according to the conditions of each street and the activities it supports.

Finally, Chapter 8 discusses the findings and limitations of the research in relation to the literature, and Chapter 9 answers the research questions and considers how the approach could be transferred to other districts, cities, and heat-vulnerable populations.

2.1 Climate Change and Heatwaves in the Mediterranean

Human activities, primarily the combustion of fossil fuels, have become the dominant driver of climate change since the Industrial Revolution (IPCC, 2022). Among its many impacts, extreme heat has become one of the most significant challenges for cities. The World Meteorological Organization defines heatwaves as periods of unusually hot weather lasting at least two to three days that have a discernible impact on human activities (WMO & WHO, 2015), and the World Health Organization notes that heatwaves "are increasing in frequency, duration, intensity and magnitude" (WHO, 2026).

These trends are particularly pronounced in the Mediterranean Basin, one of the world's most vulnerable climate change hotspots (MedECC, 2020), where heatwaves have become more frequent, prolonged, and intense since the 1980s (IPCC, 2022). Summer 2025 illustrated this: global surface temperatures were well above the 1951–1980 baseline (Figure 1.1), and large parts of southern Europe and the eastern Mediterranean experienced strong to extreme heat stress (Figure 2.1). Urban areas are particularly exposed, as their dense built environments accumulate heat more rapidly and retain it for longer, placing increasing pressure on Mediterranean cities to reduce heat exposure, particularly for vulnerable populations.

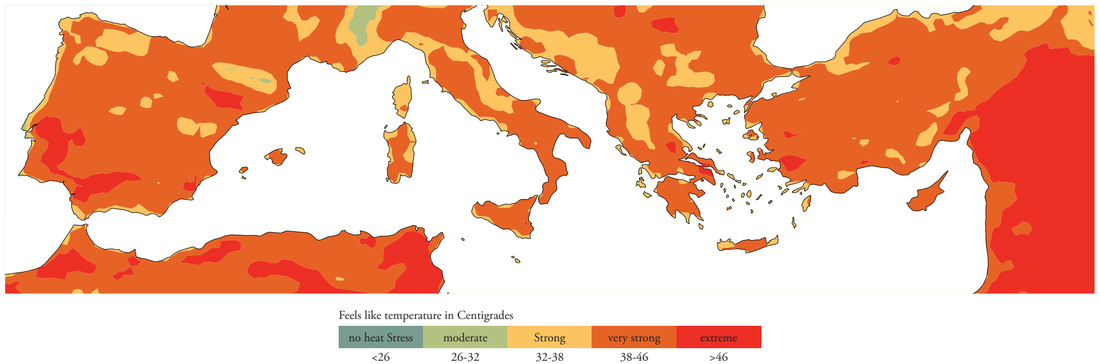


Figure 2.1. Maximum heat stress across Europe during summer 2025, expressed as the Universal Thermal Climate Index (UTCI). Source: C3S/ECMWF (2025).

2.2 The Urban Heat Island Effect and its Drivers

Cities experience an additional warming phenomenon known as the Urban Heat Island (UHI) effect, in which built-up areas record higher temperatures than their surrounding rural environments (Oke, 1982; Figure 2.2). Crucially, the UHI does not simply add to regional warming: Li and Bou-Zeid (2013) demonstrated that urban heat islands intensify during heatwaves, so that the combined impact in cities is larger than the sum of the two effects. As the IPCC has stated, "the extra warming from the urban heat island effect will exacerbate the impacts of climate change" (IPCC, 2021).

The intensity of the UHI is shaped by the interaction between regional climate and the physical and functional characteristics of cities. Three groups of drivers are commonly identified (Oke, 1982; Taha, 1997; Figure 2.3). First, urban morphology and impervious surfaces replace natural land cover with buildings, roads, and paved areas that absorb and store solar radiation during the day and release it slowly at night, while dense building configurations can limit ventilation. Second, the loss of vegetation reduces the cooling provided by shading and evapotranspiration (Taha, 1997). Third, anthropogenic heat from transport, buildings, and cooling systems adds waste heat to the urban environment, particularly in commercial centres and along transport corridors (Sailor, 2011). As the relative importance of these drivers varies with urban form and land use, the UHI is unevenly distributed within cities. Identifying the dominant drivers in a given area is therefore a prerequisite for selecting effective, context-specific mitigation measures.

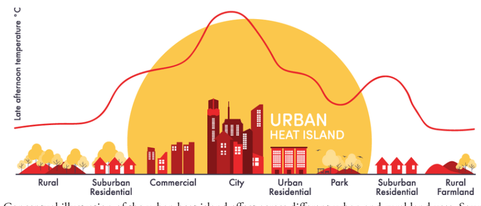


Figure 2.2. Conceptual illustration of the urban heat island effect across different urban and rural land uses. Source: Kamyar Fuladlu, in WMO (2025).

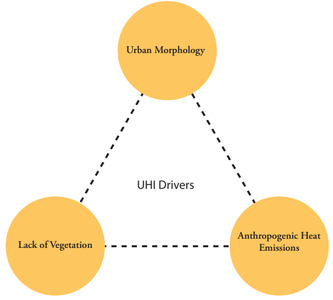


Figure 2.3. The three main groups of UHI drivers. Source: author, based on Oke (1982) and Taha (1997).

2.3 Urban Heat in Istanbul

Istanbul has been the subject of a growing body of UHI research, which confirms both the presence of the phenomenon and its close link to urban form. Using meteorological data from 30 stations between 2016 and 2022, Özçe and Erdem Kaya (2024) found that urban stations recorded monthly mean temperatures about 1 °C higher, and monthly minimum temperatures about 1.5 °C higher, than rural and suburban stations, as well as more days above 35 °C. Stations close to water bodies were less likely to exceed this threshold, indicating the moderating role of the sea.

At the surface level, Erdem Okumus and Terzi (2021) examined how urban fabric shapes the surface urban heat island in Istanbul. Their model explained approximately 71% of land surface temperature anomalies through five variables: building coverage ratio, surface-to-volume ratio, sky view factor, canyon geometry, and vegetation (NDVI). Vegetation and building coverage were the strongest contributors, while street canyon geometry had a weaker effect at the city scale. Together, these studies indicate that, in Istanbul, the loss of vegetation and the extent of built and impervious cover are central to urban heat, and that the distribution of heat varies considerably across the city's diverse urban fabric (Figure 2.4).

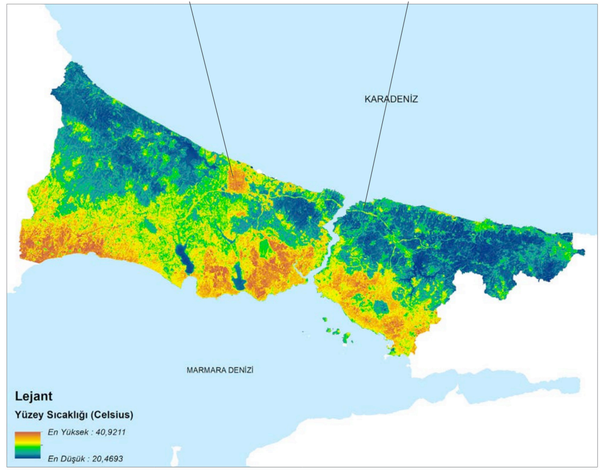


Figure 2.4. Surface temperature across Istanbul (°C), ranging from approximately 20.5 °C to 40.9 °C. Source: Kadıköy Belediyesi (2021), Kadıköy 2030 mevcut durum raporu, Map 79.

2.4 Heat Vulnerability and Outdoor Thermal Comfort

The impacts of urban heat are not experienced equally. Their severity depends not only on environmental conditions, but also on the vulnerability of the people exposed. In a review of heat and public health, Kovats and Hajat (2008) identify older age, chronic illness, and social isolation among the main risk factors for heat-related illness and mortality. Age-related changes in thermoregulation, including reduced sweating and skin blood flow, limit the body's ability to dissipate heat, while a higher prevalence of cardiovascular, respiratory, and other chronic conditions further increases sensitivity during extreme heat (WHO, 2025).

Heat vulnerability also has a spatial and everyday dimension. Exposure occurs not only where people live, but also as they walk, wait for public transport, rest, and spend time in public space. This is particularly relevant for older adults, whose mobility is closely tied to their immediate built environment: in a comprehensive review, Rosso et al. (2011) found that characteristics of the local environment, such as access to nearby destinations, pedestrian infrastructure, and places to rest, are associated with walking and mobility in later life. Uncomfortable outdoor conditions can therefore restrict access to everyday destinations and participation in public life, in addition to affecting health.

Understanding this exposure requires attention to conditions at pedestrian level. Outdoor thermal comfort depends on air temperature, mean radiant temperature, humidity, and wind speed, together with personal factors such as clothing and metabolic activity (Johansson et al., 2014). In hot conditions, mean radiant temperature, which is strongly determined by direct and reflected solar radiation, is a dominant factor, making direct sun exposure and the availability of shade key components of pedestrian comfort. Since walking, waiting, and resting involve different levels of activity and exposure, the thermal requirements of public space vary with how it is used.

2.5 NBS for Urban Cooling: Evidence and Limits

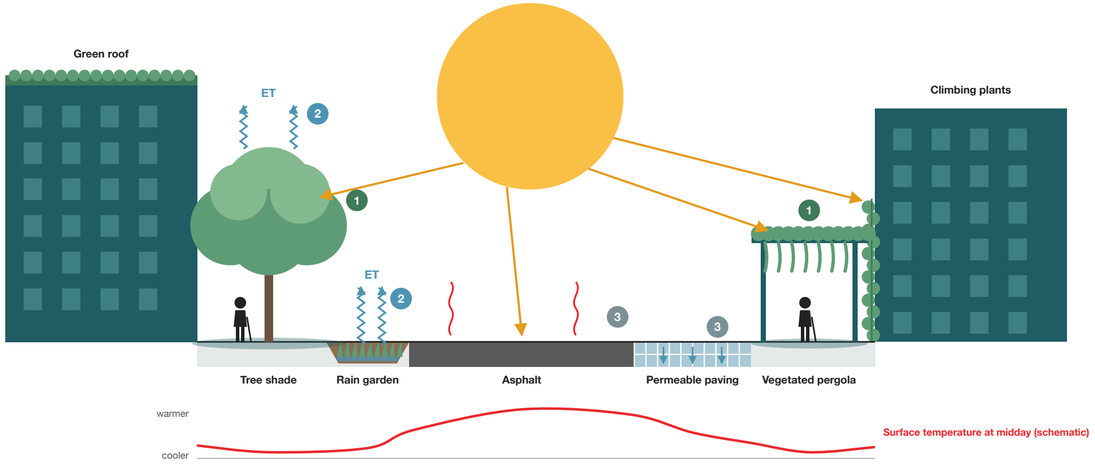
UHI mitigation strategies seek to modify the urban energy balance through two main mechanisms: increasing surface reflectivity (albedo) to reduce the absorption of solar radiation, and enhancing evapotranspiration to dissipate heat through vegetation and water (Taha, 1997). Engineered solutions such as reflective roofs and cool pavements act primarily on albedo, whereas vegetation-based interventions cool through shading and evapotranspiration while also providing environmental and social co-benefits.

Among these approaches, Nature-Based Solutions (NBS) have gained prominence in urban climate adaptation (Kabisch et al., 2016). The European Commission defines them as "solutions that are inspired and supported by nature, which are cost-effective, simultaneously provide environmental, social and economic benefits and help build resilience through locally adapted, resource-efficient and systemic interventions" (European Commission, 2025). Within urban heat mitigation, NBS reduce temperatures through three main mechanisms: shading, which intercepts solar radiation before it reaches people and surfaces; evapotranspiration, which converts energy into latent heat; and the replacement of impervious surfaces with vegetated or permeable ground, which reduces heat storage (World Bank, 2021; Žuvela-Aloise et al., 2025).

Empirical evidence supports their cooling potential, but also shows that it varies considerably. In a systematic review, Bowler et al. (2010) found that urban parks were on average around 1 °C cooler than built-up areas during the day, but noted that the evidence on how the size, type, and distribution of greenery influence cooling remained limited. Cooling also depends on quantity: Ziter et al. (2019) found that the strongest daytime cooling occurred only where tree canopy cover exceeded approximately 40% at the scale of a city block. Above all, cooling depends on setting. Comparing street trees in three street canyons, Coutts et al. (2016) found that trees were highly effective at reducing pedestrian heat stress, mainly by lowering mean radiant temperature through shade. However, their benefit increased in shallower and wider streets, whereas in deep canyons it was largely masked by the shade already provided by buildings.

Not all NBS contribute equally to pedestrian cooling. Green roofs lower roof surface temperatures and building cooling demand, but have limited effect at street level, while rain gardens and permeable surfaces cool mainly when moisture is available for evaporation (Žuvela-Aloise et al., 2025). Tree canopies can also reduce night-time cooling in enclosed streets by limiting the release of heat to the sky (Bowler et al., 2010). The same intervention can therefore perform very differently depending on where and how it is applied.

Despite this evidence, NBS research has paid more attention to their benefits than to their implementation. Wickenberg et al. (2021) similarly highlight the limited attention given to implementation in existing NBS frameworks, emphasising the need for collaboration and the co-creation of knowledge in adapting NBS to local conditions.



	1 SHADING	2 EVAPOTRANSPIRATION	3 SURFACE MODIFICATION
HOW IT COOLS	Tree canopies, vegetated pergolas and climbing plants intercept direct solar radiation before it reaches people, pavements and façades.	Plants and moist soil use solar energy to evaporate water, turning sensible heat into latent heat.	Replacing asphalt and concrete with planted, permeable or lighter surfaces reduces the solar heat the street absorbs and stores.
EFFECT FOR PEDESTRIANS	Lower mean radiant temperature, the main driver of daytime heat stress outdoors, and cooler surfaces beneath.	Cooler air around planting, parks and rain gardens.	Cooler ground by day and less heat released into the street at night.
EVIDENCE	Street trees reduce pedestrian heat stress mainly through shade (Coutts et al., 2016); the strongest daytime cooling occurs above about 40% canopy cover (Ziter et al., 2019).	Urban parks are on average about 1 °C cooler than built-up areas during the day (Bowler et al., 2010).	In Istanbul, vegetation and building coverage are the strongest drivers of land surface temperature (Erdem Okumus & Terzi, 2021).
LIMITS	Smaller benefit in deep canyons already shaded by buildings (Coutts et al., 2016); dense canopies can slow night-time cooling in enclosed streets (Bowler et al., 2010).	Depends on available water: cooling weakens when the soil dries out in long dry periods (Žuvela-Aloise et al., 2016).	Permeable surfaces and rain gardens cool mainly when moist; green roofs cool roofs rather than the street below (Žuvela-Aloise et al., 2016).

Figure 2.5. How nature-based solutions cool streets: shading, evapotranspiration and surface modification, with the evidence for and limits of each. Source: author, based on Bowler et al. (2010), Coutts et al. (2016), Erdem Okumus and Terzi (2021), Ziter et al. (2019) and Žuvela-Aloise et al. (2016).

2.6 Street Geometry and Cooling in Dense Urban Fabric

In dense urban areas, the geometry of streets is itself a major determinant of pedestrian thermal comfort. Using microclimate simulations, Ali-Toudert and Mayer (2006) showed that the ratio of building height to street width and street orientation strongly influence thermal sensation at street level, primarily through their effect on sun exposure. Shading was identified as the most important factor for summer comfort, and was best achieved by combining design elements such as building geometry, overhangs, galleries, and trees.

Field measurements confirm this in historic urban fabric. In Fez, Morocco, Johansson (2006) found that deep, narrow streets in the traditional medina were considerably more comfortable during summer days than wide streets in modern districts, as buildings shaded most of the street surface. Dense, organically grown fabric can therefore offer inherent cooling through self-shading, while wider streets, squares, and open spaces are often the most exposed.

Together with the findings of Coutts et al. (2016), these studies suggest that cooling interventions in dense fabric should be targeted: additional shade is most valuable where streets are wide and sun-exposed, while in narrow, self-shaded streets, other measures may be more appropriate or intervention may be less necessary. Since narrow streets also offer limited space for tree planting, the selection of NBS must respond to both thermal need and spatial feasibility.

2.7 Prioritising Urban Cooling: Existing Frameworks

As resources for urban greening are limited, several frameworks have been developed to guide where cooling interventions should be located. Norton et al. (2015) proposed a framework for prioritising green infrastructure that first identifies neighbourhoods where high surface temperatures coincide with vulnerable populations, and then selects appropriate types of green infrastructure according to street characteristics. Catalogues such as the World Bank's Catalogue of Nature-Based Solutions for Urban Resilience (World Bank, 2021) provide overviews of available measures and their benefits, but are not tied to specific places or users.

These approaches establish an important link between heat exposure, social vulnerability, and the selection of interventions. However, vulnerability is often defined by where people live. For groups whose exposure depends strongly on everyday movement, such as older adults walking to nearby destinations, the places that matter most may lie along the routes and destinations they use daily.

2.8 Exemplars: Cooling Networks for Everyday Life

Two programmes illustrate how urban cooling can be organised around everyday movement rather than around individual green spaces. Barcelona has developed a network of around 400 climate shelters, including libraries, civic centres, parks and courtyard gardens, that are freely accessible and intended particularly for older people, young children and people with chronic illnesses (Barcelona City Council, n.d.). According to official figures, 97% of residents live within a ten-minute walk of a shelter. However, an assessment of the network during the summer of 2023, which took into account opening hours and the slower walking speed of older adults under heat stress, found that actual access varied considerably over the day and fell to 15% of the city's older population at night (Meza et al., 2026). The study shows that proximity alone does not guarantee access, since the walk to a cool place is itself part of the heat exposure that older adults face.

In Medellín, the Green Corridors programme converted streets into continuous vegetated corridors, planting almost 880,000 trees and 2.5 million smaller plants between 2016 and 2019. Programme monitoring reports that average air temperatures at corridor locations fell from 31.6 °C to 27.1 °C, and surface temperatures from 40.5 °C to 30.2 °C, over this period (Sustainable Energy for All, 2021). Although these figures have not been independently evaluated, and Medellín's climate differs from that of Istanbul, the programme shows how linear, connected greening along streets can be implemented across a whole city.

Together, the two exemplars point towards the approach developed in this thesis: cooling organised as a connected network around everyday destinations, in which the routes between destinations matter as much as the destinations themselves.

2.9 Nature-Based Solutions in Türkiye

In Türkiye, nature-based solutions have entered urban policy mainly through climate action planning and large-scale green space programmes. The Istanbul Climate Change Action Plan (İBB, 2021) identifies rising temperatures and heat waves as a major risk for the city. Its proposed measures for heat, however, are framed primarily as tree planting and the expansion of green areas, without specifying where, in what form, or for whom these measures should be implemented. At the national level, the most visible greening programme has been the Millet Bahçeleri (Nation's Gardens), promoted since 2018 and built in cities across the country. Recent scholarship has described these gardens as large, centrally driven projects whose location and design reflect political priorities more than the everyday needs of neighbourhoods (Şahin & İnan, 2024). Their proximity has also been used in real estate marketing, contributing to what has been described as green gentrification (Aktepe, 2025). Projects of this kind add green space at the scale of the city, but they do little to change the conditions of the streets that residents walk through every day.

More integrated examples have emerged through internationally funded projects. In İzmir, the European Union's URBAN GreenUP project implemented a wider programme of nature-based solutions, including green corridors, urban forests and green pavements, vertical gardens and green façades, rain gardens, bioswales and permeable pavements (URBAN GreenUP, 2025). Its central intervention, the ecological corridor along the Peynircioğlu Stream in Mavişehir, combined the planting of around 1,150 trees with the partial removal of concrete from the stream banks and with permeable walking and cycling paths (CORDIS, 2021). The project shows that a broader range of solutions can be implemented in a Turkish city. However, its documented outcomes for heat are reported qualitatively rather than as measured reductions in temperature or thermal stress.

Academic research on nature-based solutions and urban heat in Türkiye has grown over the past decade.

At the city scale, the studies reviewed in Section 2.3 link surface heat in Istanbul above all to building coverage and the loss of vegetation (Erdem Okumus & Terzi, 2021).

At pedestrian level, Sarı et al. (2025) measured thermal conditions in Istanbul street canyons 12 m wide and lined by 18 m buildings during the extreme summer of 2023. They found that Platanus orientalis and Populus canadensis could reduce the Physiological Equivalent Temperature by 5–6 °C, while Robinia pseudoacacia reduced it by only around 1 °C. Even the most effective trees did not bring pedestrians out of extreme heat stress, and east–west streets were consistently more stressed than north–south streets.

In Elazığ, an ENVI-met study found that increasing tree density by 20% lowered summer air temperature by about 0.6 °C, and that combined green and blue infrastructure achieved about 1.5 °C. Green roofs, by contrast, had almost no effect at pedestrian height (Yilmaz et al., 2025).

At the level of decision-making, Şeker and Orak developed a Bayesian network model that uses Local Climate Zones to select suitable blue–green infrastructure for different parts of Istanbul (Şeker & Orak, 2023; Seker & Orak, 2024). The authors acknowledge that the small number of such applications in the city meant no implementation data were available to validate the model. For dense urban areas, their model ranked green roofs and urban gardens highest, solutions that act at the scale of the building and the district rather than the street.

In Kadıköy itself, Abravanel (2025) assessed nature-based stormwater management in the Acıbadem and Hasanpaşa neighbourhoods, selected for their high share of impervious surfaces, but that study addressed runoff and flooding rather than heat.

Taken together, current practice and research in Türkiye reveal two gaps. The first is a gap of scale. Policy and implementation operate mainly at the scale of the city, the district or the large park, while research has either modelled the city as a whole or tested individual elements in isolation, leaving the street largely unaddressed as a unit of intervention. The second is a gap of focus. Existing work rarely asks whose heat exposure is being reduced, or in which spaces of everyday life it occurs, even though the people most sensitive to heat, such as older adults, are exposed above all while walking, waiting and resting in the street. This thesis addresses both gaps, by taking the street as the unit of intervention and the everyday routes and destinations of older adults as the basis for deciding where cooling should be prioritised.

3. Methodology

This research adopts a multi-scalar, research-by-design methodology combining spatial and environmental analysis, demographic data, field observation, an exploratory interview, and design exploration. Kadıköy, a district on the Anatolian side of Istanbul, serves as the case study, owing to its ageing population, its diverse urban fabric, and a centre in which everyday destinations are concentrated and walking is a primary mode of movement. The process moves from district-scale identification of priority areas to neighbourhood-scale investigation of older adults' everyday destinations, and finally to street-scale environmental analysis and design. Through this process, the social context informs where cooling should be prioritised, while local spatial and environmental conditions inform how Nature-Based Solutions (NBS) can be selected and adapted. The methodology is structured into five interconnected stages.

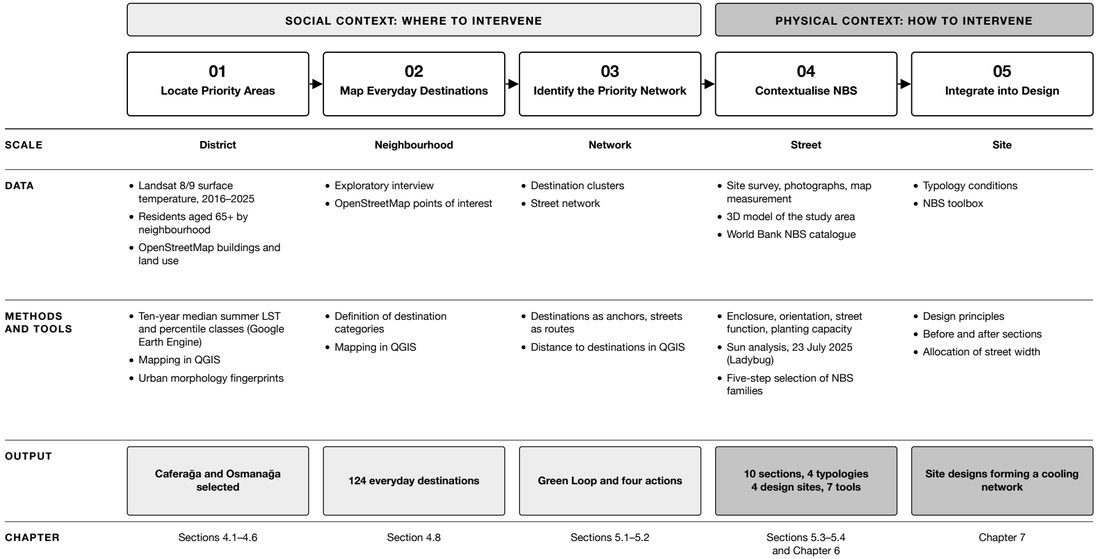


Figure 3.1. Methodology framework: the five stages of the research, showing how the social context determines where to intervene and the physical context determines how to intervene, with the scale, data, methods and tools, and output of each stage. Source: author.

01. Locate Priority Areas

The research begins by identifying the spatial distribution of surface heat and older residents across Kadıköy. Land Surface Temperature (LST) is used as an initial screening layer. LST describes the temperature of surfaces as measured by satellite thermal sensors, and differs from air temperature, which is measured above the ground and experienced by people. Although it does not directly represent pedestrian thermal comfort, LST is widely used to map surface urban heat islands, as it offers continuous spatial coverage where weather stations are sparse.

Landsat 8 and 9 imagery (USGS Collection 2) for June, July, and August between 2016 and 2025 was processed in Google Earth Engine. Cloud-affected pixels and permanent water surfaces were masked, a median summer composite was generated for each year, and the ten annual composites were combined into a ten-year median, reducing the influence of individual heat events. The result was classified using percentile thresholds as elevated (75th–90th), high (90th–95th), and very high (above 95th). Demographic data on residents aged 65 and over were obtained from the Kadıköy Municipality open-data portal and mapped by neighbourhood in QGIS.

Rather than treating all areas of high LST as equal priorities, heat and demographic patterns are interpreted alongside land use and urban form. Hotspots on large single-use sites, such as institutional campuses, sports facilities, transport infrastructure, and commercial clusters detached from residential areas, are distinguished from those within the residential and mixed-use fabric where older adults live and walk, and only the latter are considered further. Within this fabric, neighbourhoods with the smallest proportion of older residents, and those whose residents have a greater capacity to adapt to heat, are excluded. Neighbourhoods are then selected where residential streets lie within walking distance of frequently visited everyday destinations, and where a dense urban fabric constrains the space available for intervention.

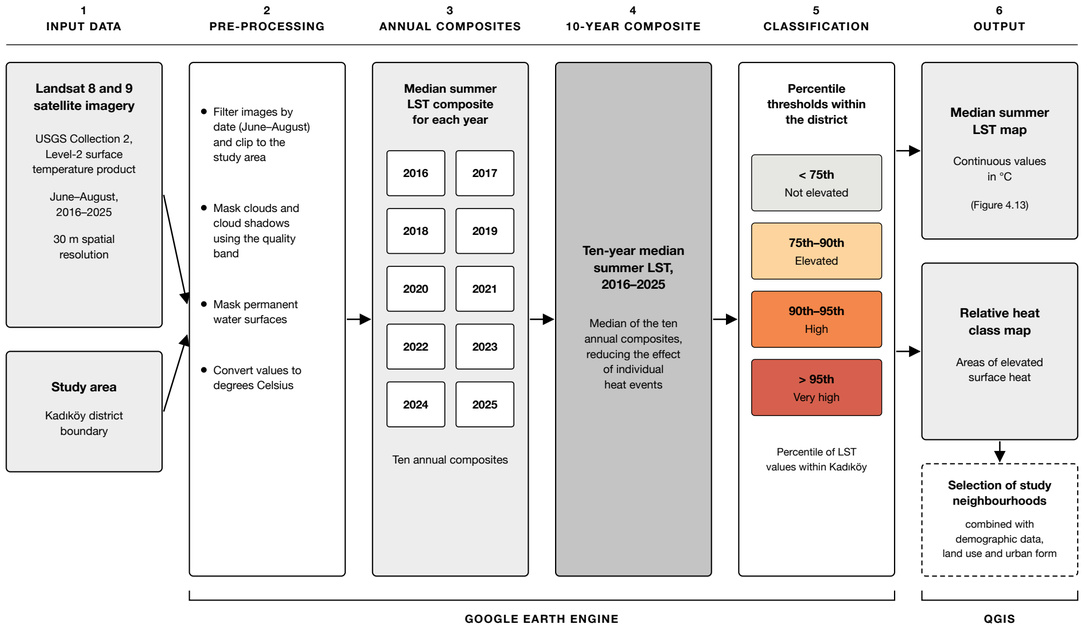


Figure 3.2. Workflow for deriving the ten-year median summer land surface temperature (LST) and relative heat classes for Kadıköy, from Landsat 8 and 9 imagery processed in Google Earth Engine to the classified map used for the selection of study neighbourhoods. Source: author; data: USGS Landsat 8 and 9 Collection 2.

02. Map Everyday Destinations

Following the selection of the study area, the research shifts from where older adults live to where they engage with the neighbourhood in everyday life. An informal exploratory interview with students from Medipol University, who had previously investigated older adults' use of public space in Kadıköy, provides insight into their everyday activities and frequently visited types of destinations. The interview is used to define destination categories, not as a representative study of behaviour.

The destination types identified through this process are extracted from OpenStreetMap and mapped in QGIS: markets and bazaars, bakeries, traditional tea houses, mosques, parks, pharmacies and healthcare facilities, together with tram stops. Mapping them shows where activities relevant to older adults are concentrated.

03. Identify the Priority Network

The spatial distribution of everyday destinations is analysed to identify clusters of activity and the main streets connecting them. Clusters of destinations are treated as anchors, and the streets linking them as priority walking routes. Together, these form a priority network that defines where cooling interventions should initially be concentrated, with the potential to extend into surrounding streets. The resulting network is presented in Chapter 5.

04. Contextualise Nature-Based Solutions

The research then examines how NBS can respond to the different spatial and environmental conditions found along the priority network. Ten representative street sections along the network were surveyed through site observation, photographs and map measurement, and described by building height, street width, enclosure (the ratio of building height to street width), orientation, street function (the allocation of street space between pedestrians, public transport and vehicles), planting capacity and existing vegetation. The sections were then grouped into typologies that share the same combination of enclosure, street function and planting capacity, so that a design developed for one section can be transferred to the others of its type.

Direct sun on pedestrian space was simulated using Ladybug Tools in Grasshopper, based on a 3D model of the study area, for 23 July 2025, selected as a representative extreme-heat summer day in Istanbul (Section 4.4). Within each typology, the section most exposed to direct sunlight was selected as a design site, so that design efforts concentrate where the heat load on pedestrians is greatest.

In parallel, an NBS toolbox was derived from the World Bank's Catalogue of Nature-Based Solutions for Urban Resilience. The fourteen NBS families in the catalogue were filtered in five steps that follow the catalogue's own categories: their function (heat regulation), the processes through which they deliver it (shade and evapotranspiration), their scale and area requirement, their suitability for high urban density, and the benefits they provide to people (heat stress risk reduction). The retained families were translated into seven street-scale tools.

05. Integrate into Design

The final stage translates the priority network and NBS toolbox into site-specific design interventions. At each design site, tools are selected according to the conditions recorded in Stage 04 and to the activities the space must support: walking, resting, or waiting. Rather than being applied uniformly, they are combined according to the opportunities and constraints of each location, and developed as components of a wider cooling network rather than as isolated projects. The effect of each design on surface conditions is described by comparing the share of the street section given to vehicles, planting and sealed surfaces before and after intervention, measured from the dimensioned sections (Section 7.8).

NBS Mechanisms Considered in This Research

As discussed in Chapter 2, NBS mitigate urban heat through three main mechanisms. Shading intercepts solar radiation before it reaches people and surfaces; as mean radiant temperature dominates daytime outdoor comfort, shade provides the most immediate benefit for pedestrians. Evapotranspiration from plants and moist soil converts solar energy into latent heat, lowering air and surface temperatures. Surface modification, replacing impervious paving with vegetated or permeable surfaces, reduces heat absorption and storage and enables evaporative cooling.

Not all NBS provide meaningful cooling at pedestrian level. Green roofs mainly cool roofs and reduce building energy demand, with limited effect in the street below. Lawns and low planting provide evaporative cooling but little shade. Rain gardens and permeable surfaces cool mainly when moisture is available, and offer limited benefit in long dry periods. Tree canopies, the most effective source of daytime shade, can reduce night-time cooling in narrow, enclosed streets. This research therefore prioritises measures that provide shade, and uses rain gardens and permeable surfaces in combination with shading rather than on their own.

4. Analysis

4.1 Istanbul

Istanbul, with almost 16 million residents, is Türkiye's largest city and one of the largest metropolitan areas in Europe (TÜİK, 2025). Located on the Bosphorus between the Black Sea and the Sea of Marmara (Figure 4.1), it has a transitional Mediterranean climate, with hot, relatively dry summers and mild, wet winters, and as part of the Mediterranean climate change hotspot, it faces increasingly frequent and intense summer heat (MedECC, 2020).

Studies of Istanbul confirm a pronounced urban heat island: urban weather stations record higher mean and minimum temperatures and more days above 35 °C than rural and suburban stations, while proximity to the sea moderates extreme heat (Ögçe & Erdem Kaya, 2024). At the surface level, vegetation and building coverage are the strongest drivers of land surface temperature across the city (Erdem Okumus & Terzi, 2021), so that the distribution of heat varies considerably across Istanbul's diverse urban fabric, from historic cores to post-war apartment districts and recent high-rise developments (Section 2.3).

At the same time, Istanbul's population is ageing. In 2025, 8.7% of its residents were aged 65 or over, below the national average of 11.1%, but older adults are unevenly distributed across the city's 39 districts (TÜİK, 2025; Figure 4.5), and their concentration in particular districts makes these areas especially relevant for heat adaptation. This chapter therefore narrows the focus from Istanbul to the district with the largest older population, Kadıköy, and from there to the neighbourhoods selected for detailed study.

Figure 4.1. Location of Türkiye (Country), Istanbul (City) and Kadıköy (District). Source: author.

4.2 Kadıköy: The District

Kadıköy is one of Istanbul's 39 districts (ilçe), located on the Anatolian side of the city along the northern coast of the Sea of Marmara, opposite the historic peninsula on the European side. Covering approximately 25.09 km² and comprising 21 neighbourhoods (mahalle), it is bordered by Üsküdar and Ataşehir to the north, Maltepe to the east, the Sea of Marmara to the south, and the Bosphorus to the west. As one of Istanbul's major residential, commercial, and transport centres, it connects the Marmara waterfront and the residential areas of the Anatolian side with the wider metropolitan network by ferry, sea bus, Marmaray, metro, metrobus, and the Kadıköy–Moda heritage tram (Figure 4.2).

Kadıköy was selected as the case study for three reasons. First, it has the largest older adult population of any district in Istanbul and one of the highest proportions of residents aged 65 or over (Section 4.3). Second, its urban fabric combines a dense, historic centre with later apartment districts and high-rise development, allowing the relationship between urban form and heat to be examined. Third, its centre is a metropolitan transport hub and an everyday destination for residents across the district, where walking is a primary mode of movement.

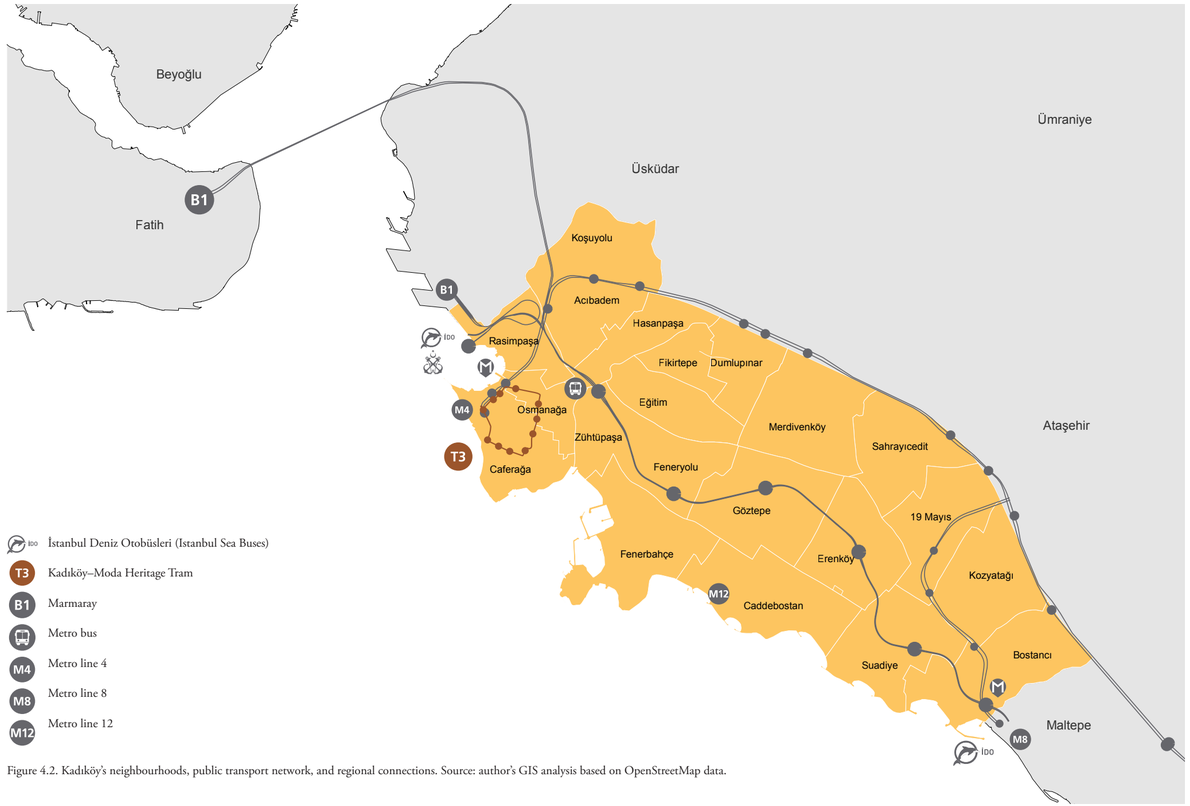


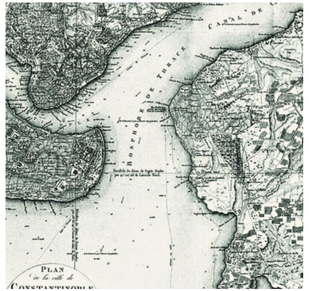
Figure 4.2. Kadıköy's neighbourhoods, public transport network, and regional connections. Source: author's GIS analysis based on OpenStreetMap data.

Historical Development of Kadıköy

Kadıköy has a long history of settlement, extending from ancient Chalcedon through the Byzantine and Ottoman periods. For much of this time, settlement was concentrated along the coast, surrounded by agricultural land, gardens, and dispersed dwellings. The historic centre, today's Caferağa and Osmanağa, developed as a compact, organically grown fabric of narrow streets shaped by walking (Figure 4.3).

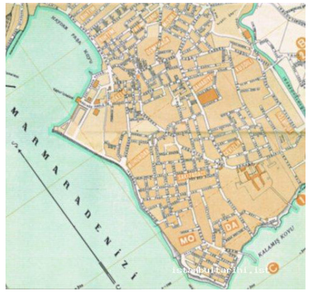
From the nineteenth century, ferry and railway connections encouraged permanent residential growth along the Marmara coast. After the establishment of the Republic, population growth and housing demand accelerated urbanisation, and former gardens and open land were progressively replaced by apartment buildings, producing a denser and more continuous urban fabric. From the 1970s, the opening of the first Bosphorus Bridge shifted movement from ferries to roads, while migration and development amnesties further increased building density (Kadıköy Belediyesi, n.d.-b; Kadıköy Kaymakamlığı, n.d.; Figure 4.4).

These successive transformations explain two conditions central to this thesis: the loss of open and green land to dense development, and a historic street network designed for pedestrians but now shared with vehicles.



(a) 1786

(b) 1918



(c) 1934

(d) 1939

Figure 4.3. Historical maps illustrating the spatial development of Kadıköy: (a) 1786, (b) 1918, (c) 1934 and (d) 1939. Source: İstanbul Tarihi (istanbultarihi.ist).

Figure 4.4. Timeline of Kadıköy's development from the first settlements to the present. Collage by author, based on Kadıköy Belediyesi (n.d.-b) and Kadıköy Kaymakamlığı (n.d.).

4.3 Population Characteristics

According to the Turkish Statistical Institute (TÜİK, 2025), Kadıköy had a population of 458,573 in 2025, approximately 2.9% of Istanbul's total population. With an area of 25.09 km², this corresponds to a density of about 18,300 residents per km², making the district one of Istanbul's most densely populated urban centres.

One of the defining demographic characteristics of Kadıköy is its ageing population. In 2025, 21.0% of the district's residents were aged 65 or over, compared with 8.7% across Istanbul and 11.1% nationally (Figure 4.6), and with 96,252 residents in this age group, Kadıköy has the largest older adult population of all districts in Istanbul (Figure 4.5). Residents aged 25–64 form the largest group, at 59.3%, while children under 15 account for only 10.5% (Figure 4.7), so the share of older adults is likely to continue to rise as this large working-age cohort ages.

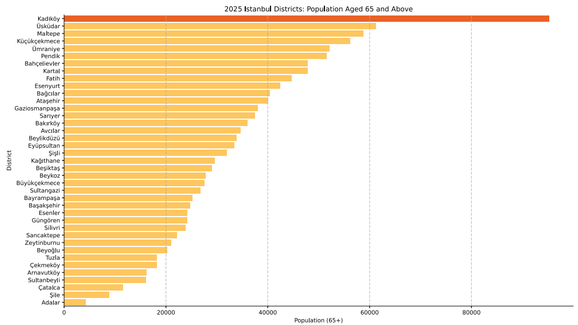


Figure 4.5. Residents aged 65+ across Istanbul's districts, 2025. Source: author, based on TÜİK (2025).

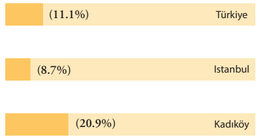


Figure 4.6. Share of residents aged 65+ in Türkiye, Istanbul, and Kadıköy, 2025. Source: author, based on TÜİK (2025).

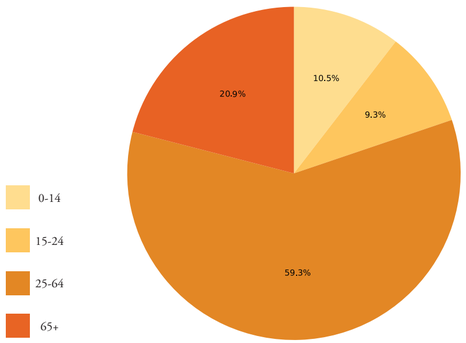
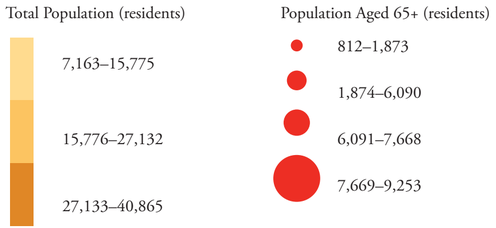


Figure 4.7. Age distribution of Kadıköy's population, 2025. Source: author, based on TÜİK (2025).

The older population is not evenly distributed across the district (Figure 4.8). The largest numbers of residents aged 65 and over live in the south-eastern neighbourhoods of Göztepe, Erenköy, Kozyatağı, Bostancı, 19 Mayıs and Suadiye, which are also among the most populous, whereas the northern and central neighbourhoods, including Fikirtepe, Dumlupınar and Hasanpaşa, have far fewer. Caferağa has a moderate number of older residents and Osmanağa one of the smallest, so the neighbourhoods selected for detailed study in Section 4.6 were not chosen for their resident population, but for the everyday destinations they contain, which serve older adults from across the district.



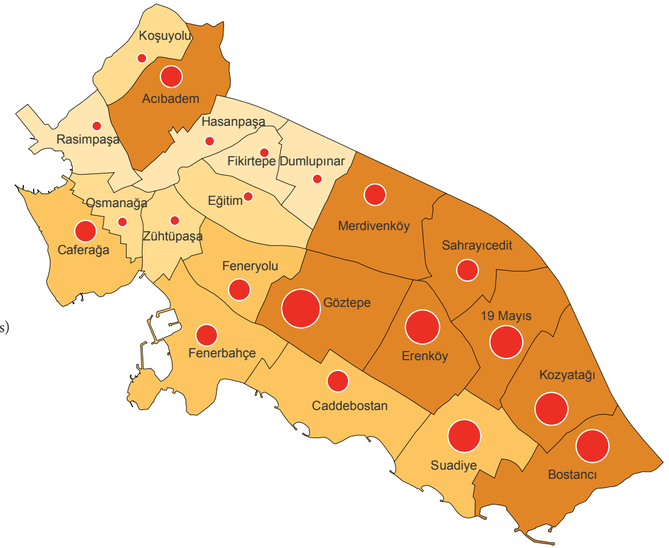


Figure 4.8. Total population and residents aged 65+ by neighbourhood in Kadıköy, 2025. Source: author, based on TÜİK (2025).

4.4 Climate and Urban Heat in Kadıköy

Climate and Summer Heat

Kadıköy has a transitional Mediterranean climate, characterised by hot, relatively dry summers and mild, wetter winters (Figure 4.9). Its coastal location along the Sea of Marmara moderates seasonal variation, while summers bring prolonged periods of high temperature, strong solar exposure and limited precipitation. The warmest conditions occur in July and August, when daytime heat can persist into the evening hours (Figure 4.10), and these months also have more days with moderate to strong winds than late spring and early summer (Figure 4.11). This regional wind, however, does not necessarily reach pedestrians, since ventilation at street level depends on the orientation and enclosure of each street, which are examined in Chapter 5.

The hourly thermal conditions of summer 2025 (Figure 4.12) show that hot conditions were concentrated between approximately noon and 6 pm from late June to late August, with very hot periods in July and August, while nights remained warm. Within this period, 23 July 2025 was selected as a representative extreme-heat day for the solar exposure analysis in Chapter 5, as it fell within the most persistent period of hot conditions.

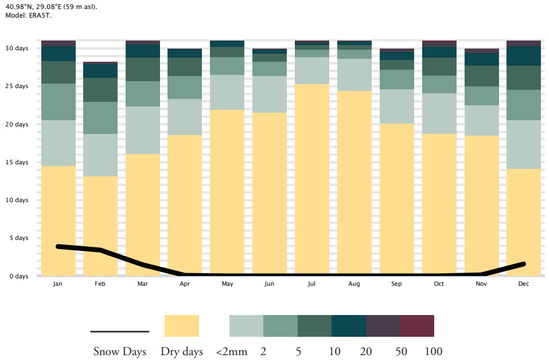


Figure 4.9. Monthly precipitation in Kadıköy. Source: Meteoblue (ERA5T model).

Although the coastal setting can provide some cooling through maritime influence, this effect diminishes within the dense urban fabric, where building configuration, street orientation, vegetation, and surface materials shape the conditions experienced at street level. Climate data describe the conditions of the district as a whole, but not how heat is distributed within it. The following analysis therefore examines surface heat across Kadıköy.

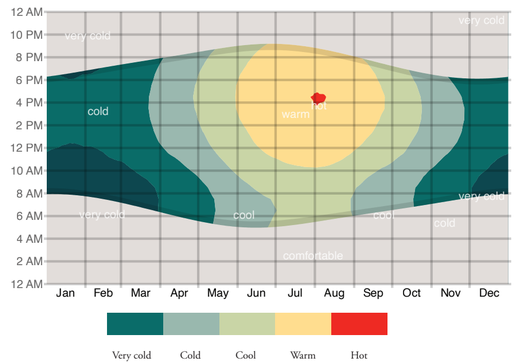


Figure 4.10. Annual pattern of hourly air temperature in Kadıköy. Source: WeatherSpark.

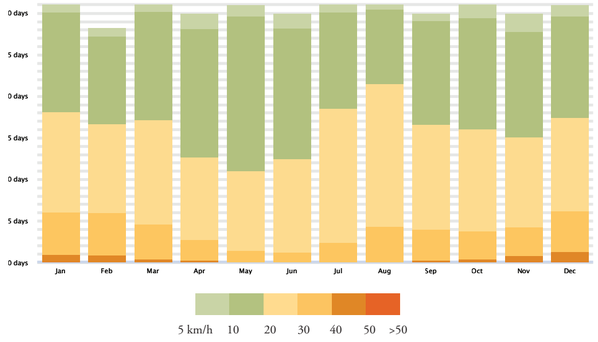


Figure 4.11. Monthly wind speed distribution in Kadıköy. Source: Meteoblue.

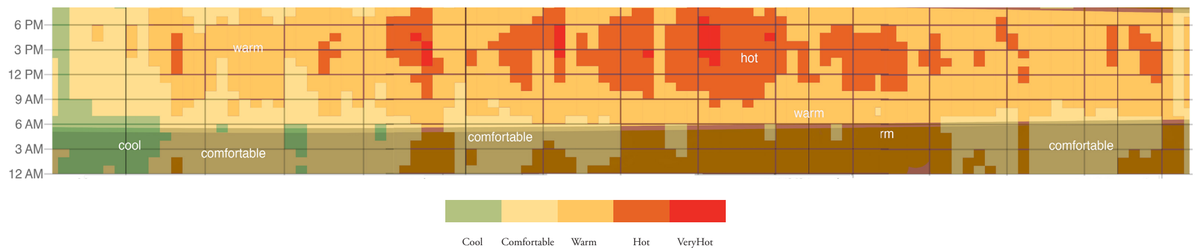


Figure 4.12. Hourly thermal conditions during summer 2025 in Kadıköy. Source: WeatherSpark.

Surface Heat across the District

To identify where heat accumulates within the district, Land Surface Temperature (LST) was analysed across Kadıköy. As explained in Chapter 3, LST describes the temperature of the ground, roofs and other surfaces as measured by satellite thermal sensors, rather than the air temperature that people experience, and on summer days dark, impervious surfaces such as asphalt and roofing can become far hotter than the surrounding air, while vegetated and shaded surfaces remain cooler. LST therefore indicates where surfaces absorb and retain the most heat, though not directly how comfortable a street is for pedestrians, which is examined at street level in Chapter 5 through solar exposure analysis. It is used here because it provides consistent and spatially continuous data across the whole district, allowing areas of elevated surface heat to be compared.

The analysis is based on a ten-year median of summer LST (June–August, 2016–2025) derived from Landsat 8 and 9 imagery, which represents recurring summer patterns rather than individual heat events (see Figure 3.2 for the processing method). The resulting values (Figure 4.13a) were classified relative to the district as elevated (75th–90th percentile), high (90th–95th) and very high (above the 95th percentile), as shown in Figure 4.13b.

The results show that surface heat is unevenly distributed across Kadıköy. The highest surface temperatures occur in the north-west of the district around Haydarpaşa port and rail yards, along its northern edge following the D-100 highway and railway corridor, and in Fikirtepe, where large-scale redevelopment has replaced much of the former fabric. Despite the district's proximity to the Sea of Marmara, elevated temperatures also remain within parts of the inner urban fabric, including the historic centre around Caferağa and Osmanağa (Figure 4.13b). The coolest areas lie along the coastline and in the residential neighbourhoods of the south-east, where gardens and street trees are more common. Areas of elevated heat coincide with extensive impervious surfaces and limited vegetation, which absorb and retain solar heat and reduce evaporative cooling, and these patterns provide the basis for the examination of urban morphologies in Section 4.5 and the neighbourhood selection in Section 4.6.

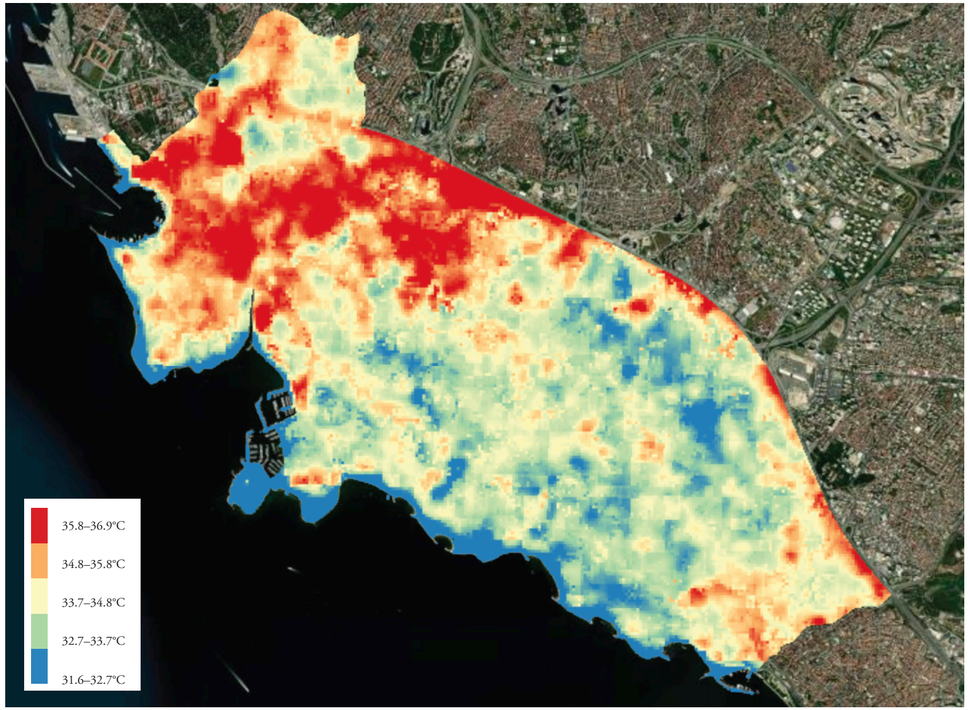


Figure 4.13a. Ten-year median summer land surface temperature (LST) across Kadıköy, June–August 2016–2025. Source: author's analysis in Google Earth Engine; data: USGS Landsat 8 and 9 Collection 2, Level-2.

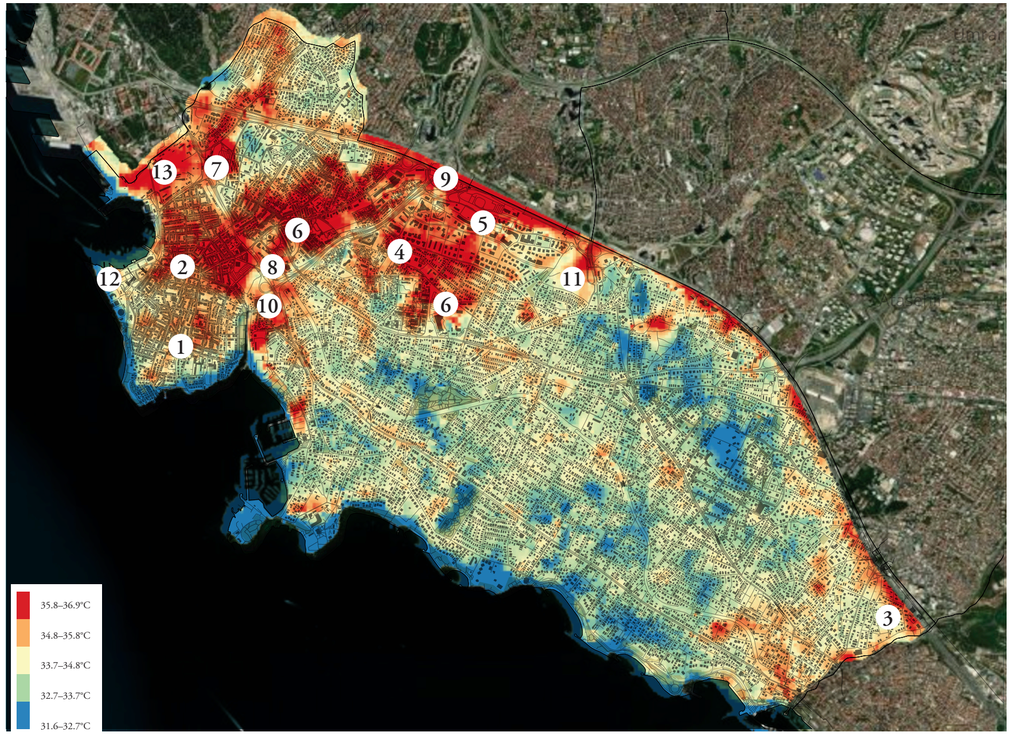


Figure 4.13b. Locations of the thirteen morphology samples shown in Figure 4.14. Source: author's analysis in Google Earth Engine and QGIS; data: USGS Landsat 8 and 9 Collection 2, Level-2.tion 2, Level-2.

4.5 Urban Morphologies of Heat

To understand why surface heat concentrates where it does, the hotspots identified in the LST analysis were examined in relation to the urban form in which they occur. Thirteen urban morphologies were identified through visual interpretation of the LST maps overlaid with satellite imagery and OpenStreetMap building footprints, and a representative sample of each was drawn as a figure-ground fingerprint at the same scale (Figures 4.13b and 4.14). The morphologies fall into two groups: the residential and mixed-use fabric in which older adults live and walk, and large single-use sites. The planned medium-rise fabric of the south-east was included as a point of comparison, since it houses the largest older population but shows comparatively lower surface temperatures. Each morphology was then characterised by its physical form, its dominant UHI drivers, drawing on the literature reviewed in Chapter 2, its constraints for intervention and its relevance to the everyday mobility of older adults (Table 4.1). The approach is comparable to Local Climate Zone classification, which groups urban areas by the form and land cover that shape their thermal behaviour (Stewart & Oke, 2012), but differs from it in also assessing each morphology from the perspective of the people who use it.

A. Residential and mixed-use fabric

1.Historic compact residential fabric (Caferağa)

2. Traditional commercial streets (Çarşı, Bahariye, Moda)(mixed-use fabric)

3.Planned medium-rise residential fabric (Kozyatağı)

4. Informal dense residential fabric (old Fikirtepe)

5. High-rise residential fabric (Dumlupınar)

B. Large single-use sites

6.Institutional Campuses

7.Large commercial complexes

8.Transport hubs

9.Transport corridors

10.Sport Complexes

11.Industrial and brownfield areas

12.Surface parking areas

13.Harbour and port areas

Figure 4.14. Urban fingerprints of Kadıköy: figure-ground samples of the thirteen urban morphologies, shown at the same scale. Numbers refer to the sample locations in Figure 4.13b and the rows of Table 4.1. Source: author, based on OpenStreetMap data.

Table 4.1

Table 4.1. Urban morphologies identified in the LST analysis: characteristics, heat drivers, constraints, relevance to older adults' everyday mobility, and decision. Source: author, based on the literature reviewed in Chapter 2.

Urban morphology	Representative examples	Morphological characteristics	Dominant UHI drivers	Design constraints	Relevance to older adults' everyday mobility	Decision
A. Residential and mixed-use fabric						
1 Historic compact residential fabric	Caferağa, Osmanağa, Rasimpaşa, Yeldeğirmeni	Attached buildings (4–7 storeys), compact blocks, organic street network, narrow streets, high plot coverage	Extensive impervious surfaces, restricted ventilation, limited evapotranspiration, high thermal mass	Very limited public space, heritage-sensitive interventions, underground utilities	High: dense housing within walking distance of everyday destinations	**Included**
2 Traditional commercial streets	Çarşı, Bahariye, Moda	Attached buildings with ground-floor shops, continuous street frontage, narrow streets, high pedestrian activity	Heat-absorbing façades, impervious pavements, anthropogenic heat, restricted airflow	Limited planting space, heavy pedestrian movement	High: concentration of everyday destinations	**Included**
3 Planned medium-rise residential fabric	Kozyatağı, Erenköy, Caddebostan, Göztepe, Suadiye	Detached or semi-detached apartment buildings (4–6 storeys), regular street layout, medium-to-wide streets, private gardens	Impervious pavements, discontinuous tree canopy, road infrastructure	On-street parking demand, fragmented green spaces	Moderate: largest older population, but lower surface heat, private gardens and more continuous tree canopy	Excluded
4 Informal dense residential fabric	Old Fikirtepe, Dumlupınar	Irregular street network, compact residential blocks, narrow streets, limited public open space	High building density, impervious surfaces, restricted airflow, low evapotranspiration	Limited planting space, fragmented ownership, narrow streets	Low: few older residents	Excluded
5 High-rise residential fabric	New Fikirtepe, Dumlupınar	Residential towers within superblocks, internal roads, parking areas, landscaped open spaces	Large roof surfaces, impervious open spaces, altered wind flow, heat storage	Privately managed open space, high development density	Low: few older residents; privately managed open space	Excluded

Table 4.1 Continued

Urban morphology	Representative examples	Morphological characteristics	Dominant UHI drivers	Design constraints	Relevance to older adults' everyday mobility	Decision
B. Large single-use sites						
6 Institutional campuses	Marmara University, Göztepe Hospital	Large building footprints within extensive plots, courtyards, schoolyards, parking areas	Large roof surfaces, paved courtyards, fragmented vegetation, anthropogenic heat	Operational requirements, emergency access, circulation needs	Low: single-use; rarely part of everyday walking	Excluded
7 Large commercial complexes	Tepe Nautilus, Indoor Bazaar	Large building footprint, extensive parking areas, wide setbacks, large roof surfaces	Roof heat gain, asphalt parking lots, low vegetation, anthropogenic heat	Vehicle access requirements, extensive paved surfaces	Low: mainly accessed by car	Excluded
8 Transport hubs	Kadıköy, Söğütlüçeşme, Bostancı	Transport terminals, paved plazas, platforms, station forecourts, park-and-ride facilities	Asphalt and concrete surfaces, anthropogenic heat, high pedestrian and vehicle density	Continuous operation, intensive circulation, limited planting areas	Indirect: origins and ends of everyday trips	Excluded as design sites; used as anchors
9 Transport corridors	D-100, major junctions	Multi-lane roads, highways, arterial streets, large intersections	Asphalt heat storage, vehicle emissions, anthropogenic heat, limited vegetation	Traffic safety, underground utilities, restricted planting zones	Low: hostile to walking	Excluded
10 Sports complexes	Fenerbahçe Stadium, Dereağzı facilities	Large open sports fields, spectator areas, medium-sized buildings, artificial surfaces	High solar exposure, artificial turf, exposed hardscape, limited tree canopy	Functional sports requirements, unobstructed playing areas	Low: single-use; occasional visits	Excluded
11 Industrial and brownfield areas	Haydarpaşa rail yard, Merdivenköy	Warehouses, vacant industrial land, construction sites, exposed ground	Concrete and bare soil heat absorption, impervious surfaces, lack of vegetation	Contamination, redevelopment phases, ownership complexity	None: no everyday use	Excluded
12 Surface parking areas	Nautilus parking, bus garage, fuel stations	Large open paved areas, vehicle circulation, minimal shading	Asphalt heat storage, solar exposure, vehicle emissions	Parking demand, circulation requirements	Low as destinations; potential space for reallocation	Excluded as design sites
13 Harbour and port areas	Haydarpaşa Port	Port infrastructure, quays, warehouses, loading areas, concrete pavements	Extensive concrete surfaces, impervious materials, anthropogenic activity, limited vegetation	Port operations, security restrictions, saline environment	None: restricted access	Excluded

Key Findings

The analysis shows that surface heat in Kadıköy is associated less with a single land use than with recurring urban morphologies, each combining heat drivers in a different way. Three findings shape the selection of the study area.

First, the most intense surface heat occurs on large single-use sites, such as transport corridors, the port and brownfield land, sports complexes, large commercial complexes, and institutional campuses. Here, extensive impervious surfaces, large roofs, and anthropogenic heat dominate. Although thermally significant, these sites rarely form part of older adults' everyday walking environment.

Second, within the residential fabric, heat varies with morphology. Informal dense and high-rise residential fabric shows elevated surface heat but houses comparatively few older adults, while the planned medium-rise fabric of the south-eastern neighbourhoods, which houses the largest older population, benefits from private gardens and more continuous tree canopy and shows comparatively lower surface temperatures.

Third, the historic compact fabric and traditional commercial streets of Caferağa and Osmanağa combine elevated surface heat with the densest concentration of everyday destinations, and with the most constrained space for intervention, owing to narrow streets, heritage sensitivities, and underground utilities. These findings form the basis of the selection described in the following section (Figure 4.15).

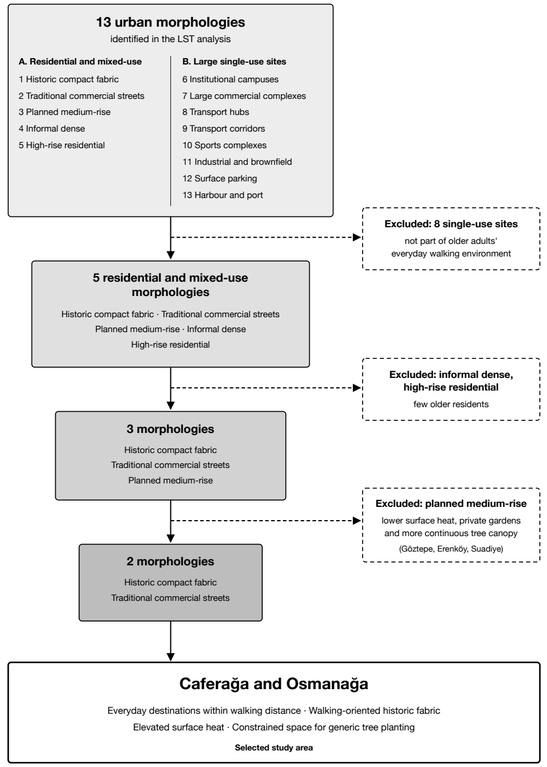


Figure 4.15. Narrowing the thirteen urban morphologies to the selected study area. Source: author.

4.6 Selection of Caferağa and Osmanağa

As shown in Section 4.5, large single-use sites were excluded because they rarely form part of older adults' everyday walking environment, informal dense and high-rise residential fabric because it houses comparatively few older adults, and the planned medium-rise fabric of the south-east because it combines lower surface heat with private gardens and a more continuous tree canopy (Table 4.1; Figure 4.15).

Caferağa and Osmanağa were selected because they combine three conditions. First, as Kadıköy's commercial and transport centre, they concentrate markets, shops, services, cafés and places of worship, together with the main ferry, tram, metro and bus connections, and their commercial core directly adjoins predominantly residential streets (Figure 4.17c). They were therefore selected not for their resident population, which in Osmanağa is comparatively small, but for the everyday destinations they contain, which serve older adults from Caferağa and the surrounding neighbourhoods alike.

Second, their organically grown, pedestrian-scaled street network means that walking remains central to everyday movement, although most streets are now shared with vehicles (Section 4.2). Third, they show elevated surface heat within the inner urban fabric (Figure 4.16), together with extensive impervious surfaces and limited vegetation, while their narrow streets constrain the space available for intervention and limit the feasibility of generic tree planting (Section 4.7). This makes them both a priority area for cooling and a demanding test case for context-specific NBS.

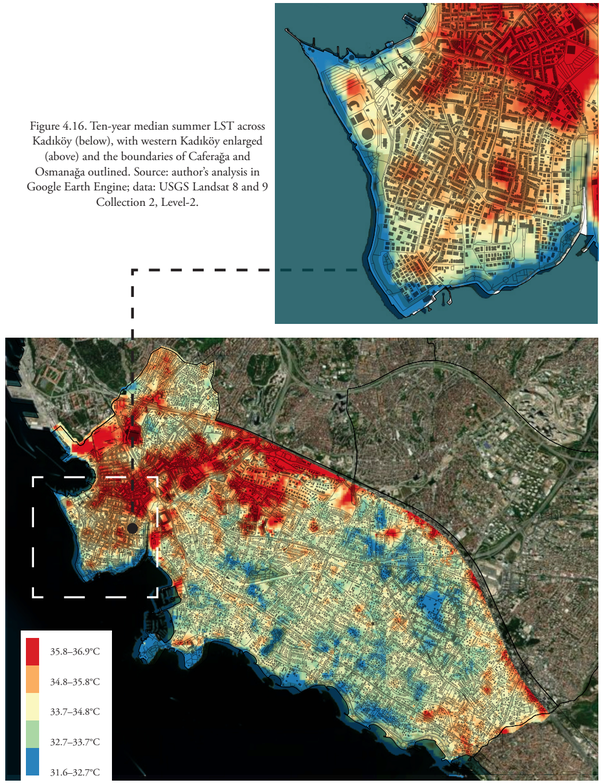


Figure 4.16. Ten-year median summer LST across Kadıköy (below), with western Kadıköy enlarged (above) and the boundaries of Caferağa and Osmanağa outlined. Source: author's analysis in Google Earth Engine; data: USGS Landsat 8 and 9 Collection 2, Level-2.

4.7 Existing Urban Conditions

Built form. Caferağa and Osmanağa are characterised by a compact urban fabric of closely spaced buildings and limited open space (Figure 4.17a). Building heights are relatively consistent, with most of the built fabric ranging between 12 and 25 metres (Figure 4.17b), and combined with the narrow width of many inner streets, this produces deep street canyons in which buildings shade much of the street surface for large parts of the day. The main streets and open spaces, by contrast, are wider and more exposed to direct sunlight. As discussed in Section 2.6, this distinction is important for cooling, since additional shade is most valuable in wide, exposed streets, while narrow, self-shaded streets may require less intervention, a difference that the solar analysis in Sections 5.3–5.4 examines street by street.

Land use and mobility. Residential and commercial uses are closely integrated. Commercial activities are concentrated along the main streets, while residential uses extend throughout the surrounding fabric, creating a mixed-use environment in which housing, retail, services, and everyday activities occur in close proximity. Street space is allocated in different ways across the area: some streets are pedestrian-only, others are shared between pedestrians and vehicles, and several main streets also accommodate the tram, buses, and cycling (Figure 4.17c). On-street and off-street parking occurs throughout the fabric.

a.Figure Ground | b.Building heights

3.Street Network and Mobility | 4.Existing Parks and Parking lots

Figure 4.17. Urban conditions in Caferağa and Osmanağa: (a) figure-ground, (b) building heights, (c) street network and mobility, (d) parks and parking. Source: author, based on OpenStreetMap data.

Vegetation and open space. Green spaces are limited and unevenly distributed. Larger green areas are located mainly at the edges of the study area, while smaller spaces are scattered within the neighbourhoods (Figure 4.18d). Street trees are present along some streets but form a discontinuous canopy.

Impervious surfaces. Impervious surfaces such as asphalt, concrete, paving, and roofs absorb solar radiation during the day and release it slowly at night, while preventing evaporation from the ground. They are therefore one of the main drivers of the urban heat island (Section 2.2), and, together with vegetation, among the strongest predictors of land surface temperature in Istanbul (Section 2.3).

Impervious surface coverage was not measured across the whole study area, as no available land-cover data distinguishes sealed from permeable ground at this scale. It was instead assessed at street level, as the share of each design section occupied by sealed surfaces before and after intervention (Section 7.6). In the surveyed streets, pavements, carriageways and tram tracks are almost entirely sealed, and unsealed ground is limited to tree pits and the edges of parks.

Figure 4.18. Street-level conditions in Caferağa and Osmanağa. Photographs: author, February 2026; deciduous street trees are shown without leaves.

4.8 Everyday Destinations

Following the selection of Caferağa and Osmanağa, the analysis shifts from where older adults live to where they engage with the neighbourhood in everyday life. As described in Chapter 3, the destination categories were defined through an informal exploratory interview with students from Medipol University who had previously studied older adults' use of public space in Kadıköy, and the interview was used to define these categories rather than as a representative study of older adults' behaviour.

Based on these categories, the corresponding locations were mapped in QGIS using OpenStreetMap data (Figure 4.19). In total, 124 destinations were identified across the two neighbourhoods: 39 pharmacies, 35 markets and bazaars, 19 bakeries, 15 healthcare facilities, 8 traditional teahouses, 6 mosques and 2 parks, while tram stops are shown for reference, as they mark the start and end of many everyday trips.

The destinations are not evenly distributed. They are concentrated in the north of the study area, around the commercial core and along a small number of main streets leading south towards Moda, while the inner residential streets and the southern part of the area contain comparatively few, and the two parks lie along the coast at the edges of the area. This pattern suggests that cooling for older adults can be concentrated on a limited network of streets rather than spread across the whole area, which is the starting point for the strategy developed in Chapter 5.

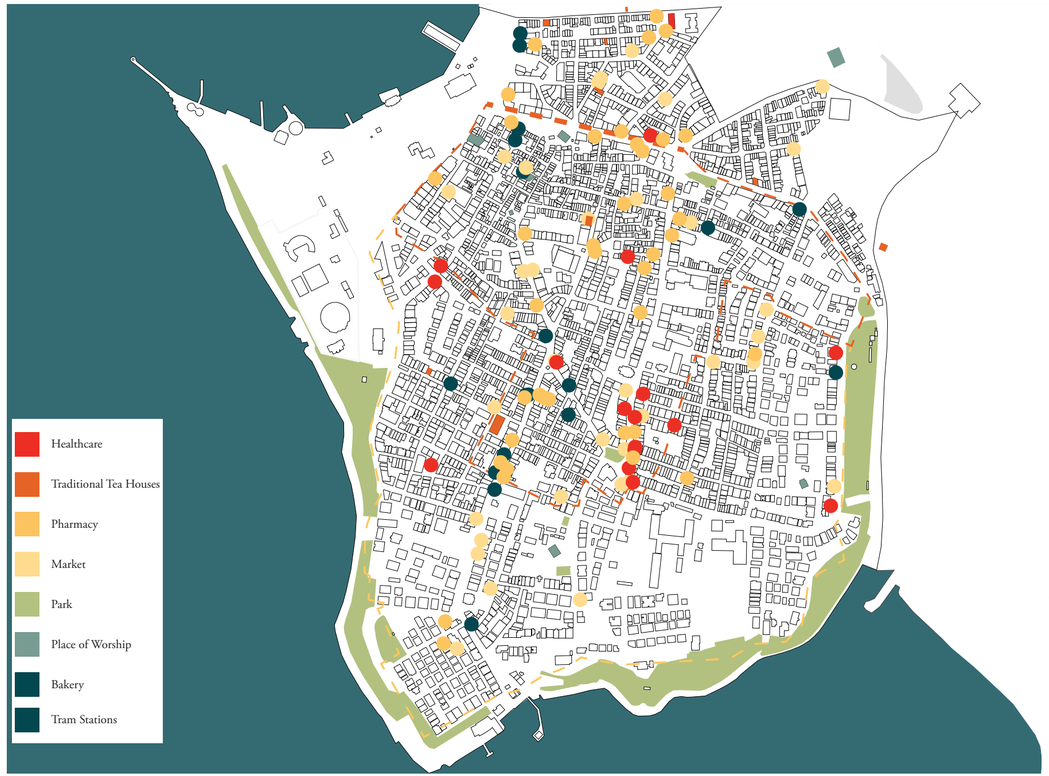


Figure 4.19. Everyday destinations relevant to older adults in Caferağa and Osmanağa. Source: author's GIS analysis based on OpenStreetMap data.

4.9 Synthesis: SWOT

The SWOT analysis synthesises the findings of the environmental and social analyses (Figure 4.20). Whereas the preceding sections describe existing conditions, the SWOT interprets them in relation to the aim of this thesis. Strengths and weaknesses describe internal conditions of Caferağa and Osmanağa that support or hinder cooling and everyday mobility. Opportunities and threats describe the potential for intervention and the external pressures that may limit it. The SWOT thereby forms the bridge between the analysis and the strategy developed in Chapter 5.

Strengths. The two neighbourhoods offer conditions that already support everyday mobility. Mixed residential and commercial uses and a concentration of everyday destinations mean that daily needs can be met on foot, while a connected street network and strong public transport accessibility make walking a practical mode of movement. Existing parks and green spaces, although located mainly at the edges of the area, provide starting points for a wider green network.

Weaknesses. At the same time, the area is thermally stressed. Elevated surface temperatures, extensive impervious surfaces, and a limited and discontinuous tree canopy leave many streets without adequate shade. The heat vulnerability of older adults means these conditions can restrict their mobility precisely in the spaces they depend on, while limited space for greening constrains the measures that can be introduced, particularly in narrow streets.

Opportunities. These conditions also suggest where and how intervention can be most effective. Everyday destinations can act as anchors around which cooling is concentrated, and shaded routes can connect them with each other and with existing green spaces. Where space allows, the street-tree canopy can be expanded; where it does not, small-scale NBS can be integrated. Parking areas represent the largest reserve of reallocatable space.

Threats. Rising summer temperatures and more frequent extreme heat will intensify existing pressures. Competition for limited public space, together with vehicle and parking pressure, may hinder the reallocation of street space. Adaptation policy focused on generic tree planting may overlook measures suited to constrained streets, while drought and water scarcity may limit the irrigation that vegetation requires.

Taken together, the SWOT indicates that the priority is not to green the area uniformly, but to concentrate cooling where everyday destinations and walking routes coincide with heat exposure, and to select measures that fit within the spatial constraints of each street. These implications inform the strategy in Chapter 5.

Strengths

- Mixed residential and commercial uses
- Concentration of everyday destinations
- Walkable and connected street network
- Existing parks and green spaces
- Strong public transport accessibility

Weaknesses

- Elevated surface temperatures
- Limited and discontinuous tree canopy
- Insufficient street shade
- Extensive impervious surfaces
- Limited space for greening
- Higher heat vulnerability of older adults

Opportunities

- Everyday destinations as cooling anchors
- Expansion of street-tree canopy
- Conversion of parking space to greenery
- Integration of small-scale NBS
- Connection of green spaces through shaded routes

Threats

- Increasing summer temperatures
- More frequent extreme heat
- Vehicle and parking pressure
- Competition for limited public space
- Generic, tree-focused adaptation policy
- Drought and water scarcity limiting irrigation

Figure 4.20. SWOT analysis of Caferağa and Osmanağa in relation to heat and everyday mobility. Source: author.

5. Strategy

The analysis in Chapter 4 showed that heat exposure in Caferağa and Osmanağa is not uniform, since it varies from street to street according to building height, street width, orientation and vegetation, and according to how street space is shared between pedestrians, vehicles and the tram. This chapter translates these findings into a spatial strategy. It first sets out the four actions through which the strategy is developed, from individual destinations to the wider neighbourhood (5.1), and then defines the Green Loop that results from connecting these destinations along Bahariye Caddesi, Moda Caddesi and Söğütlüçeşme Caddesi (5.2). It then describes ten street sections along the loop and groups them into four typologies (5.3), and finally selects the design sites in which the toolbox is applied in Chapter 7 (5.4).

Figure 5.1. Schematic implementation plan informed by the identified key destinations. Source: Author.

5.1 Four Actions

The strategy is developed through four actions, applied in sequence from individual destinations to the wider neighbourhood.

Anchor: Identify the everyday destinations most frequently used by older adults, such as markets, mosques, clinics, tram stops, squares and kıraathanes, and establish them as thermal comfort anchors where shaded spaces for resting and waiting are provided first.

Cool: Improve the immediate surroundings of each anchor through context-specific nature-based solutions, increasing shade, vegetation and permeable surfaces so that arriving, waiting and resting take place in thermally comfortable conditions.

Connect: Link the anchors through continuous shaded walking routes, prioritising the street segments most exposed to direct sunlight, so that the walk between destinations does not involve long unshaded stretches.

Expand: Extend the network from the loop into the adjoining residential streets over time, applying the same typology-based logic, so that thermal comfort and green connectivity gradually reach the doorstep.

Together the four actions establish an order of priority. Shade comes first where older adults wait and rest, then along the routes between destinations, and finally in the residential streets that lead into them.

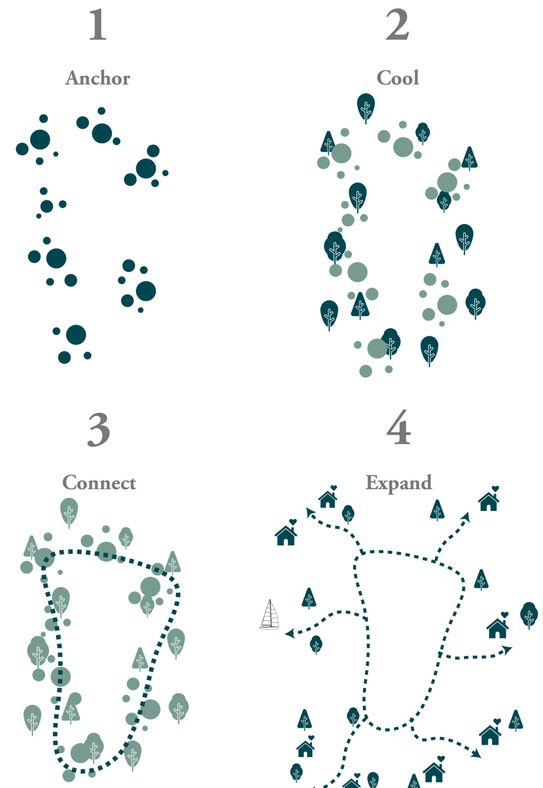


Figure 5.2. The four actions of the strategy: Anchor, Cool, Connect and Expand. Source: author.

5.2 The Green Loop: From Anchors to a Cooling Network

Older adults do not experience heat as an average condition of the neighbourhood. They experience it as a sequence of spaces they walk through, wait in and rest in on the way to their everyday destinations, and their time outdoors is concentrated along the routes between home and markets, places of worship, healthcare facilities and public spaces. The strategy therefore focuses on these routes rather than spreading interventions evenly across the neighbourhood, where they would be less likely to coincide with the spaces older adults actually use.

The destinations mapped in Section 4.8 are concentrated along three main streets, Bahariye Caddesi, Moda Caddesi and Söğütlüçeşme Caddesi, which together form a continuous circuit through the two neighbourhoods (Figure 5.3). A loop was chosen over a single corridor for three reasons: it connects destinations in both directions, it offers alternative routes of similar length, and it allows a daily walk to begin and end at home without retracing the same exposed street.

Of the 122 everyday destinations mapped in Section 4.8, excluding the two parks, approximately two-thirds (82) lie within 100 m of the loop, a walk of about two minutes at an older adult's pace, and almost 90% (109) within 200 m (Table 5.1).

The loop follows the main streets rather than the quieter side streets, because these are where destinations, tram and bus stops and daily activity are concentrated, so that older adults have to walk along them whatever the thermal conditions. Historically shaped for pedestrians but now shared with vehicles and on-street parking, several of them are among the most exposed streets in the study area, which makes them both the most problematic and the most consequential places to intervene, since a shaded route along them benefits the largest number of daily trips.

The Green Loop is the first stage of a cooling network rather than the network itself: it establishes a continuous, shaded spine between the most frequently visited destinations, from which the network can gradually extend into the adjoining residential streets. Side streets are addressed at the strategic level rather than as design sites in this thesis, since many of them are narrow and deeply enclosed and therefore already shaded by their buildings for much of the day, and the typologies defined in Section 5.3 provide the means of transferring the toolbox to them in later phases.

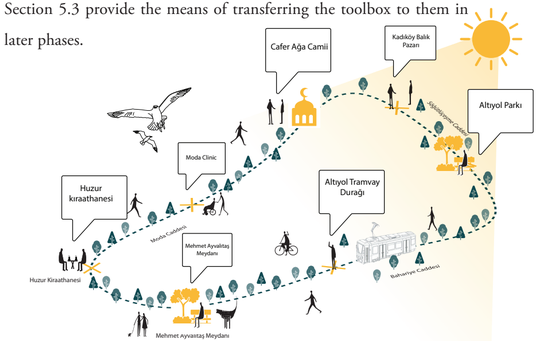


Figure 5.3. The Green Loop: everyday destinations as anchors and the shaded routes along Bahariye, Moda and Söğütlüçeşme Caddesi that connect them (schematic). Source: author.

Table 5.1. Everyday destinations by distance to the Green Loop.

Destination type	Total	Within 100 m of the loop	Within 200 m of the loop	Share by distance
Pharmacies	39	24 (62%)	36 (92%)	
Markets	35	25 (71%)	32 (91%)	
Bakeries	19	14 (74%)	18 (95%)	
Healthcare facilities	15	13 (87%)	14 (93%)	
Traditional teahouses	8	4 (50%)	7 (88%)	
Mosques	6	2 (33%)	2 (33%)	
All destinations	**122**	**82 (67%)**	**109 (89%)**	

within 100 m 100–200 m more than 200 m

5.3 The Built Environment along the Green Loop

How hot a pedestrian feels depends on the weather, on personal factors such as age, health and clothing, and on the built environment. The built environment decides how much solar radiation reaches the street, how much heat its surfaces store, and how freely air moves through it. Among the built-environment factors, urban morphology, vegetation, surface materials, density, ventilation, traffic and anthropogenic heat all add to the intensity of urban heat (Chapter 2). Climatic and personal factors are broadly similar along the loop. This section therefore focuses on what differs from one street to the next and what design can change: the geometry of the street, the way its space is shared, and how much vegetation and unsealed ground it has. In Istanbul, building coverage and vegetation have been found to be the strongest predictors of land surface temperature anomalies (Erdem Okumus & Terzi, 2021).

The streets along the loop were first described by their built form. Building height decides how much of the street is shaded by the surrounding buildings, while street width sets the space available for interventions. Width therefore decides the scale of nature-based solutions that can be introduced, from climbing plants in the narrowest streets to rows of large trees in the widest. Together, height and width give the height-to-width ratio (H/W), which describes how enclosed a street is. Deep street canyons receive less direct sunlight during the day but release stored heat slowly at night, whereas open streets stay exposed for most of the day. In this thesis, enclosure is classified as low (H/W < 0.5), moderate (0.5–1.4), high (1.5–2.0) or very high (> 2.0).

Orientation decides when, and for how long, each side of the street is in direct sunlight. In Istanbul, east–west streets have been found to experience higher thermal stress than north–south streets, because one side stays exposed to the sun for most of the day (Sarı et al., 2025).

Each street was then described by how it is used. Street function describes how the street is shared between pedestrians, the tram, cyclists and vehicles, including on-street parking. It shows which parts of the street could be given over to vegetation and shade without blocking essential access. Planting capacity describes the space available for planting both below and above ground, taking into account pavement width, underground utilities, tram infrastructure and access. It decides whether trees can be planted in the ground or in planter boxes, or whether only climbing plants and pergolas are possible.

Finally, solar exposure was simulated with Ladybug for 23 July 2025 and rated as high, medium or low according to how much of the pedestrian space of each street remains in direct sunlight during the day. Based on these criteria, ten sections along the Green Loop, A-A to J-J, were surveyed (Figures 5.5–5.14; Table 5.2) and grouped into four typologies (Table 5.3), and the way in which the design sites were selected from these typologies is explained in Section 5.4 and summarised in Figure 5.4.

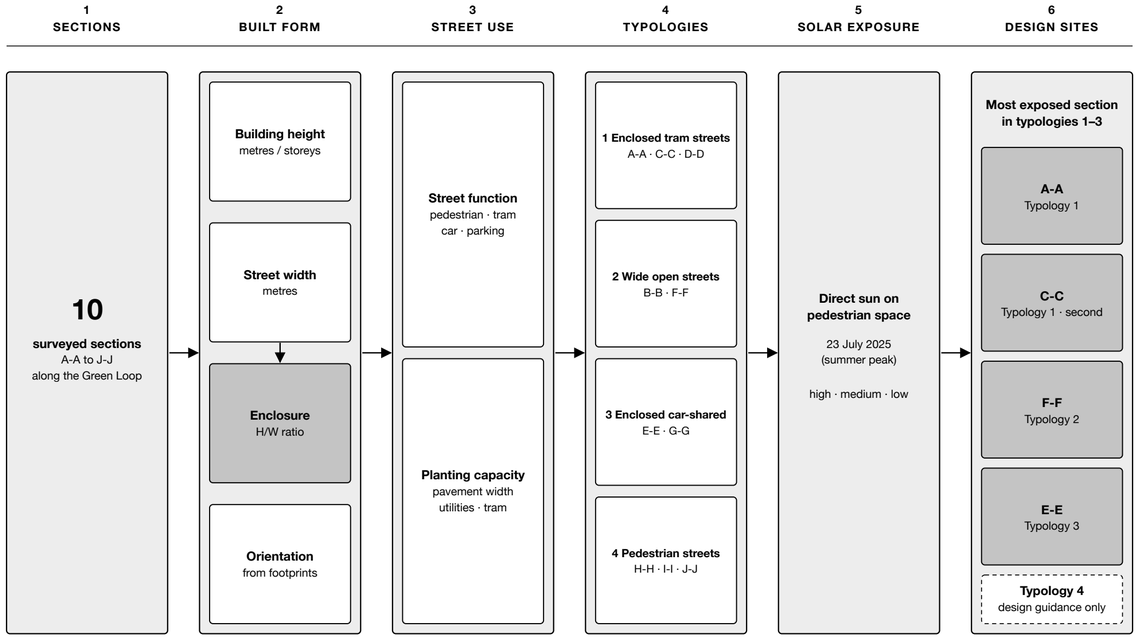


Figure 5.4. Workflow for surveying the sections, grouping them into typologies and selecting the design sites. Source: author.

Section A-A: Pedestrian–Tram Corridor

Building height: 21 m / 7 storeys

Street width: 12.5 m

Street orientation: N–S

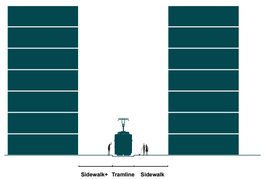
Urban enclosure: High (H/W 1.7)

Solar exposure (23 July): High

Street function: Pedestrian + tram

Planting capacity: Moderate

Existing condition: Street trees + generous pedestrian edges



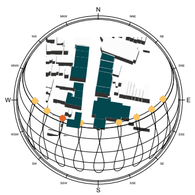


Figure 5.5. Section A-A, pedestrian–tram corridor: section, sun path on 23 July 2025 (dots mark hourly sun positions) and street view. Source: author; sun path in Ladybug.

Section B-B: Park-Edge Street

Building height: 18 m / average 6 storeys

Street width: 40 m

Street orientation: N-S

Urban enclosure: Low (H/W 0.45)

Solar Exposure: Medium (northern side)

Street function: Pedestrian + car

Planting capacity: Moderate–high

Existing condition: Existing park + generous pedestrian edges

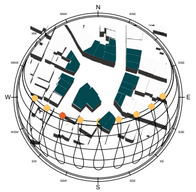


Figure 5.6. Section B-B, park-edge street: section, sun path on 23 July 2025 (dots mark hourly sun positions) and street view. Source: author; sun path in Ladybug.

Section C-C: Mixed Tram Street

Building height: 15 m / approx. 5 storeys

Street width: 10 m

Street orientation: NE–SW

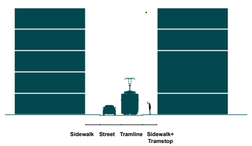
Urban enclosure: High (H/W 1.5)

Street function: Pedestrian + tram + car

Planting capacity: Low–moderate

Existing condition: Street trees + narrow pedestrian edges

Solar exposure (23 July): Medium–high



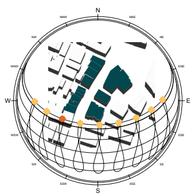


Figure 5.7. Section C-C, mixed tram street: section, sun path on 23 July 2025 (dots mark hourly sun positions) and street view. Source: author; sun path in Ladybug.

Section D-D: Narrow Tram Street

Building height: 15 m / approx. 5 storeys

Street width: 6.5 m

Street orientation: E–W

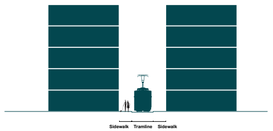
Urban enclosure: Very high (H/W 2.3)

Street function: Pedestrian + tram

Planting capacity: Low

Existing condition: Street trees + very narrow pedestrian edges

Solar exposure (23 July): Low



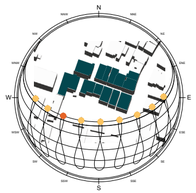


Figure 5.8. Section D-D, narrow tram street: section, sun path on 23 July 2025 (dots mark hourly sun positions) and street view. Source: author; sun path in Ladybug.

Section E-E: Parking-Dominated Street

Building height: 18 m / approx. 6 storeys

Street width: 13.5 m

Street orientation: NNE–SSW

Urban enclosure: Moderate (H/W 1.3)

Street function: Pedestrian + car + on-street parking

Planting capacity: Moderate

Existing condition: No vegetation + moderate pedestrian edges + extensive on-street parking

Solar exposure (23 July): High

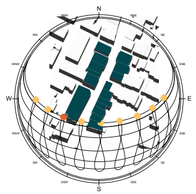


Figure 5.9. Section E-E, parking-dominated street: section, sun path on 23 July 2025 (dots mark hourly sun positions) and street view. Source: author; sun path in Ladybug.

Section F-F: Boulevard Street

Building height: 18 m / approx. 6 storeys

Street width: 19.2 m

Street orientation: N–S

Urban enclosure: Moderate (H/W 0.94)

Street function: Pedestrian + car + on-street parking

Planting capacity: Moderate–high

Existing condition: Boulevard trees + moderate pedestrian edges + extensive on-street parking

Solar exposure (23 July): High

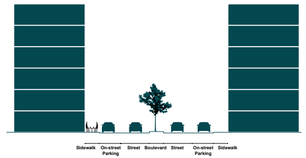


Figure 5.10. Section F-F, boulevard street: section, sun path on 23 July 2025 (dots mark hourly sun positions) and street view. Source: author; sun path in Ladybug.

Section G-G: Commercial Street (Moda Caddesi)

Building height: 18 m / approx. 6 storeys

Street width: 10 m

Street orientation: NNW–SSE

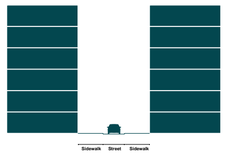
Urban enclosure: High (H/W 1.8)

Street function: Pedestrian + car

Planting capacity: Moderate

Existing condition: No vegetation + generous pedestrian edges

Solar exposure (23 July): High



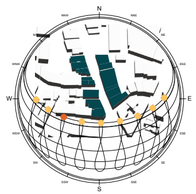


Figure 5.11. Section G-G, commercial street (Moda Caddesi): section, sun path on 23 July 2025 (dots mark hourly sun positions) and street view. Source: author; sun path in Ladybug.

Section H-H: Shared Market Street

Building height: 12 m / approx. 4 storeys

Street width: 11 m

Street orientation: NE–SW

Urban enclosure: Moderate (H/W 1.1)

Street function: Pedestrian-dominant + limited car access

Planting capacity: Moderate

Existing condition: No vegetation + generous pedestrian edges

Solar exposure (23 July): High

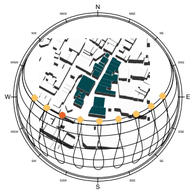


Figure 5.12. Section H-H, shared market street: section, sun path on 23 July 2025 (dots mark hourly sun positions) and street view. Source: author; sun path in Ladybug.

Section I-I: Open Pedestrian Square

Building height: 6 m / approx. 2 storeys

Street width: 18 m

Street orientation: Open square (no dominant axis)

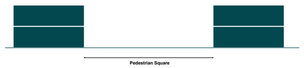
Urban enclosure: Low (H/W 0.3)

Street function: Pedestrian-dominant + occasional vehicle access

Planting capacity: Moderate–high

Existing condition: Street trees + generous pedestrian edges + public square

Solar exposure (23 July): High



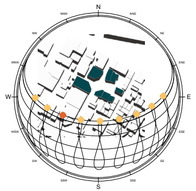


Figure 5.13. Section I-I, open pedestrian square: section, sun path on 23 July 2025 (dots mark hourly sun positions) and street view. Source: author; sun path in Ladybug.

Section J-J: Narrow Pedestrian Street

Building height: 9 m / approx. 3 storeys

Street width: 6 m

Street orientation: NW–SE

Urban enclosure: High (H/W 1.5)

Street function: Pedestrian

Planting capacity: Low

Existing condition: No vegetation + constrained pedestrian space

Solar exposure (23 July): Low

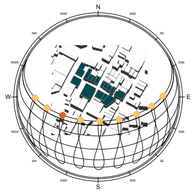


Figure 5.14. Section J-J, narrow pedestrian street: section, sun path on 23 July 2025 (dots mark hourly sun positions) and street view. Source: author; sun path in Ladybug.

5.4 Selection of Design Sites

The ten sections surveyed along the Green Loop (Table 5.2) were grouped into four typologies according to their enclosure, street function and planting capacity (Table 5.3). Each typology represents a distinct combination of constraints and opportunities for cooling, ranging from enclosed tram streets, where tram infrastructure limits in-ground planting, to wide open streets, where comfort depends on the continuity of the tree canopy.

Within Typologies 1 to 3, the section whose pedestrian space was most exposed to direct sunlight on 23 July 2025 was selected as the design site, since this is where older adults walking, waiting and resting experience the greatest heat load; E-E was chosen over G-G, which was equally exposed, because it has no vegetation and more street space available for reallocation. Typology 1 is developed through a second section, C-C, because it covers the widest range of street widths, from 6.5 m to 12.5 m, and the comparison between a 12.5 m and a 10 m tram street shows how the same response is adjusted when a traffic lane runs beside the tram. Sections that remain shaded for most of the day by their enclosure and orientation, D-D and J-J, were not prioritised for intervention.

This selection produces four design sites: Sections A-A and C-C in Typology 1, F-F in Typology 2 and E-E in Typology 3. The pedestrian streets and squares of Typology 4, whose most exposed section is H-H, are not developed as a detailed design within the scope of this thesis; since they are already largely free of traffic, cooling there does not depend on the reallocation of street space that the other typologies require, and the design guidance in Table 5.3 indicates how the toolbox would be applied. Together, the four designs show how the toolbox responds to the conditions of three typologies, and how a design developed for one section can be transferred to the other sections of its typology.

Table 5.2. Characteristics, solar exposure and selection of the ten surveyed sections. Source: author, based on site survey, map measurement and Ladybug simulation.

Section	Building height	Street width	H/W	Enclosure	Orientation	Street function	Existing vegetation	Planting capacity	Solar exposure (23 July)	Typology	Decision
A-A Pedestrian–tram corridor	**21 m (7 st.)**	**12.5 m**	**1.7**	**High**	**N–S**	**Pedestrian + tram**	**Street trees**	**Moderate**	**High**	**1**	**Selected**
B-B Park-edge street	18 m (6 st.)	40 m	0.45	Low	N–S	Pedestrian + car	Park	Moderate–high	Medium (northern side)	2	Not selected
C-C Mixed tram street	**15 m (5 st.)**	**10 m**	**1.5**	**High**	**NE–SW**	**Pedestrian + tram + car**	**Street trees**	**Low–moderate**	**Medium–high**	**1**	**Selected (second tram street)**
D-D Narrow tram street	15 m (5 st.)	6.5 m	2.3	Very high	E–W	Pedestrian + tram	Street trees	Low	Low	1	Not selected
E-E Parking-dominated street	**18 m (6 st.)**	**13.5 m**	**1.3**	**Moderate**	**NNE–SSW**	**Pedestrian + car, parking**	**None**	**Moderate**	**High**	**3**	**Selected**
F-F Boulevard street	**18 m (6 st.)**	**19.2 m**	**0.94**	**Moderate**	**N–S**	**Pedestrian + car, parking**	**Boulevard trees**	**Moderate–high**	**High**	**2**	**Selected**
G-G Commercial street (Moda Cd.)	18 m (6 st.)	10 m	1.8	High	NNW–SSE	Pedestrian + car	None	Moderate	High	3	Not selected
H-H Shared market street	12 m (4 st.)	11 m	1.1	Moderate	NE–SW	Pedestrian-dominant, limited cars	None	Moderate	High	4	Not designed (guidance only)
I-I Open pedestrian square	6 m (2 st.)	18 m	0.3	Low	Open square	Pedestrian-dominant, occasional vehicles	Street trees	Moderate–high	High	4	Not selected
J-J Narrow pedestrian street	9 m (3 st.)	6 m	1.5	High	NW–SE	Pedestrian	None	Low	Low	4	Not selected

Table 5.3. The four street typologies along the Green Loop and their design response. Source: author.

	1 Enclosed tram streets	2 Wide open streets	3 Enclosed car-shared streets	4 Pedestrian streets and squares
Sections and solar exposure	A-A high · C-C medium–high · D-D low	F-F high · B-B medium (northern side)	E-E high · G-G high	H-H high · I-I high · J-J low
Enclosure (H/W)	High–very high (1.5–2.3)	Low–moderate (0.45–0.94)	Moderate–high (1.3–1.8)	Low–high (0.3–1.5)
Street function	Pedestrian + tram (+ car)	Pedestrian + car, parking	Pedestrian + car, parking	Pedestrian-dominant
Planting capacity	Low–moderate	Moderate–high	Moderate	Low to moderate–high
Existing vegetation	Street trees	Park and boulevard trees	None	None to scattered trees
Heat-related condition	Buildings shade the street for part of the day; tram tracks, overhead lines and narrow pavements limit in-ground planting	Little building shade; comfort depends on the continuity of the tree canopy	Street edges occupied by cars and parking; no vegetation; extensive asphalt	Intensive pedestrian use; narrow streets leave little room for trees, squares are open to the sun
Design response	Trees in planter boxes; a tree with bench where the ground allows; vegetated pergolas at tram stops; climbing plants on façades	Complete the canopy: trees and rain gardens in former parking lanes; benches under trees; climbing plants	Reallocate parking to planted strips with trees and benches; climbing plants; permeable surfaces	Vegetated pergolas along frontages; trees in planter boxes with benches, or in the ground in squares; climbing plants; permeable surfaces
Design site	**A-A, C-C**	**F-F**	**E-E**	**Guidance only**

6. NBS Toolbox

The NBS toolbox translates the cooling mechanisms discussed in Chapter 2, namely shading, evapotranspiration and surface modification, into seven spatial tools that can be selected and combined along the Green Loop. Rather than prescribing a single standard response such as tree planting, the toolbox links each tool to the street conditions under which it can be implemented and to the everyday activities of older adults it supports, walking, resting and waiting (Figure 6.1), so that the same cooling objective can be pursued by different means in a wide boulevard and in a narrow historic street. Section 6.1 explains how the tools were derived from the World Bank's catalogue of nature-based solutions, Section 6.2 describes the seven tools and the conditions under which each is used, and Section 6.3 positions the toolbox in relation to existing nature-based solutions in Türkiye.

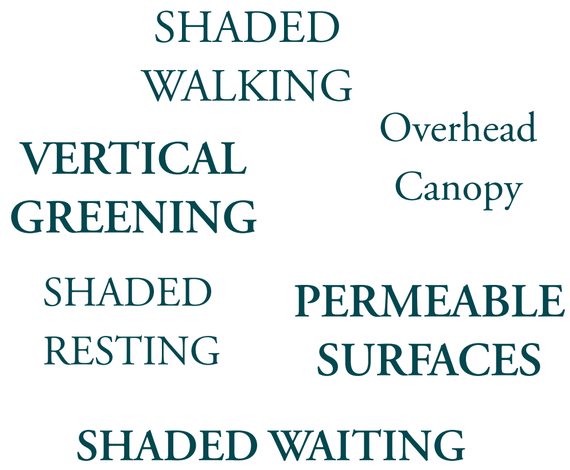


Figure 6.1. The everyday activities and spatial modes addressed by the toolbox. Source: author.

6.1 Origin and Selection of the Tools

The tools were derived from the Catalogue of Nature-Based Solutions for Urban Resilience (World Bank, 2021), which organises nature-based solutions into fourteen families. Each family is described through the processes it relies on, such as shade, evapotranspiration, infiltration or water storage; the functions these processes perform in regulating hazards, such as heat regulation or pluvial flood regulation; and the benefits they deliver to people, such as heat stress risk reduction, human health and social interaction. Each family is also assigned the scales at which it can be applied (river basin, city or neighbourhood) and a set of suitability considerations, including the area it requires and the urban density for which it is suitable.

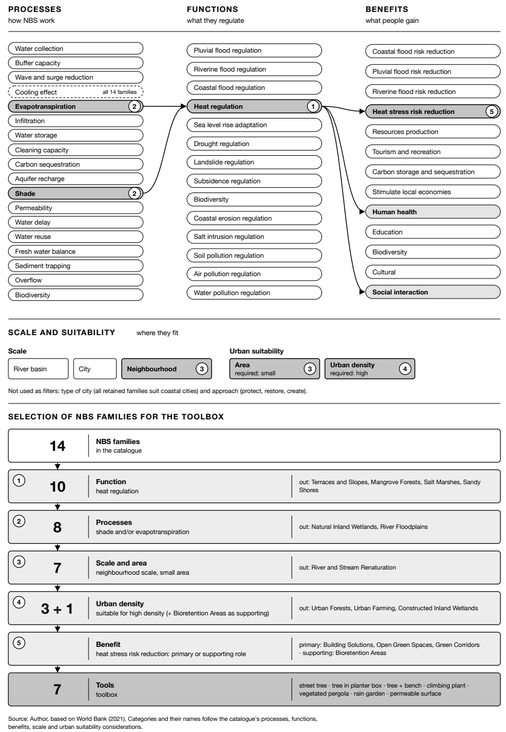


Figure 6.2. The structure of the World Bank catalogue and the five steps used to select NBS families for the toolbox. Source: author, based on World Bank (2021).

The selection followed the structure of the catalogue (Figure 6.2; Table 6.1). First, at the level of functions, only families associated with heat regulation were retained (ten families). Second, at the level of processes, only those families that regulate heat through shade and/or evapotranspiration were retained, since these processes act directly on the conditions experienced by pedestrians. Natural inland wetlands and river floodplains, which regulate heat mainly through open water and riparian forest, were therefore excluded (eight families). The more general "cooling effect", listed for all fourteen families, was not used to distinguish between them. Third, only families applicable at the neighbourhood scale and requiring a small area were retained, so that they could be integrated into existing streets and public spaces (seven families). Fourth, only families rated suitable for high urban density were retained, reflecting the dense historic fabric of Caferağa and Osmanağa: Building Solutions, Open Green Spaces and Green Corridors. Finally, the benefits were used to assign roles. These three families rate heat stress risk reduction as a relevant benefit and form the primary cooling families. Bioretention Areas, which fail the density criterion and rate heat stress reduction as least relevant, were re-admitted as a supporting family, because the catalogue describes rain gardens, bioswales and permeable pavements for streetscapes and because they retain the soil moisture that sustains street trees. From these families, seven street-scale tools were derived: four correspond directly to techniques described in the catalogue (street tree canopies, ground-based green façades, rain gardens and permeable pavements), while three were adapted by the author to the constraints of the study area, the tree in a planter box from street tree canopies, the tree with bench from pocket parks, and the vegetated pergola from green façades.

Table 6.1. Selection of NBS families from the World Bank catalogue for the toolbox. Source: author, based on World Bank (2021).

NBS FAMILY	1 FUNCTION	2 PROCESSES		3 SCALE AND AREA		4 URBAN DENSITY	5 BENEFIT	OUTCOME: TOOLBOX
	Heat regulation	Shade	Evapo-transpiration	Neighbour-hood scale	Area	Urban density	Heat stress risk reduction	
Urban Forests	●	●	●	●	**S–XL**	L–M	●	Out at 4: low–medium density
Terraces and Slopes	○	○	●	●	M–L	L–M	○	Out at 1: no heat regulation
River and Stream Renaturation	●	●	●	●	M–XL	**L–H**	○	Out at 3: medium–extra-large area
Building Solutions	●	●	●	●	**S–M**	**L–H**	●	**Climbing plant · Vegetated pergola***
Open Green Spaces	●	●	●	●	**S–L**	**L–H**	●	**Tree + bench***
Green Corridors	●	●	●	●	**S–L**	**L–H**	●	**Street tree · Tree in planter box***
Urban Farming	●	○	●	●	**S–M**	L–M	○	Out at 4: low–medium density
Bioretention Areas	●	●	●	●	**S–M**	L–M	○	**Rain garden · Permeable surface (supporting)**
Natural Inland Wetlands	●	○	○	○	M–XL	site	○	Out at 2: no shade or evapotranspiration
Constructed Inland Wetlands	●	○	●	●	**S–M**	L–M	●	Out at 4: low–medium density
River Floodplains	●	○	○	○	M–L	L–M	●	Out at 2: no shade or evapotranspiration
Mangrove Forests	○	○	○	○	L–XL	L–M	○	Out at 1: no heat regulation
Salt Marshes	○	○	○	○	L–XL	L–M	○	Out at 1: no heat regulation
Sandy Shores	○	○	○	○	L–XL	L–M	○	Out at 1: no heat regulation
Families retained	**10 of 14**	**8 of 10**		**7 of 8**		**3 of 7**	**role**	**3 primary + 1 supporting families: 7 tools**

6.2 The Seven Tools

The seven tools translate the selected families into interventions that fit within the existing streets of Caferağa and Osmanağa. Each tool is illustrated as a street section above and a plan below (Figure 6.3), showing its position relative to buildings, pavements and parked cars, and the process through which it reduces heat: shade cast on pedestrians and surfaces, evapotranspiration (ET) cooling the surrounding air, or infiltration retaining water in the soil. The first five tools are primary cooling tools that provide shade, while the last two are supporting tools that sustain planting and reduce the heat stored in paved surfaces, and Table 6.2 summarises the mechanism, activity, spatial requirement and main typologies of each.

Street tree (Green Corridors): A tree planted in the ground along the pavement, providing the most effective shade and evapotranspiration where pavement width and underground conditions allow sufficient root volume.

Tree + bench (Open Green Spaces, adapted): A street tree with seating in its shade, creating resting points at intervals that allow older adults to pause during longer walks.

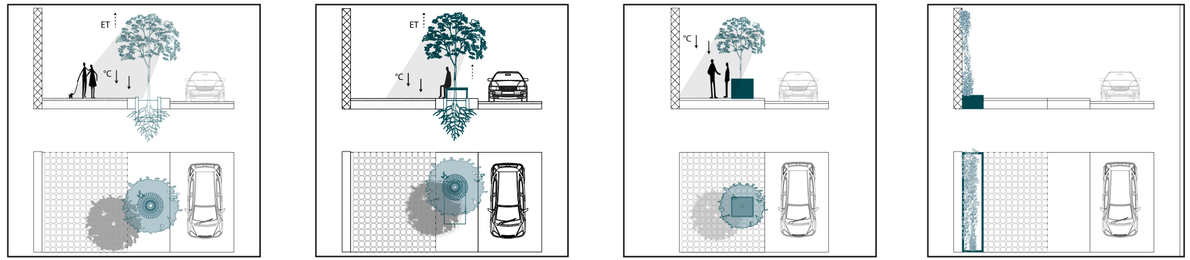
Tree in planter box (Green Corridors, adapted): A tree in a raised container where utilities, tram infrastructure or basements prevent in-ground planting; its canopy is smaller and it requires regular irrigation.

Climbing plant (Building Solutions): Vegetation on façades or walls that shades building surfaces and reduces the heat radiated towards pedestrians, requiring only a narrow planting strip.

Vegetated pergola (Building Solutions, adapted): A light overhead structure covered with climbing plants, providing shade where trees cannot be planted, such as tram stops and narrow streets.

Rain garden (Bioretention Areas): A planted depression that collects runoff from adjacent paving, retaining water in the soil that sustains nearby trees; suited to former parking bays.

Permeable surface (Bioretention Areas): Paving that allows water to infiltrate and stores less heat than asphalt, used in parking bays, squares and low-traffic areas.



1. Street Tree | 2.Tree + Bench | 3. Tree in planter box | 4. Climbing Plant

5. Vegetated Pergola | 6. Rain garden | 7. Permeable Surface

Figure 6.3. The seven tools of the NBS toolbox, each shown in section (above) and plan (below). ET: evapotranspiration · °C: local cooling · grey area: shade cast at midday. Source: author, adapted from

Table 6.2

Table 6.2. The NBS toolbox: cooling mechanism, activity supported, spatial requirement and main typologies of each tool. Source: author, adapted from World Bank (2021).

Tool	Main cooling mechanism	Activity supported	Spatial requirement	Main typologies (Table 5.3)
Primary cooling tools				
1 Street tree	Shading; evapotranspiration	Walking	In-ground pit; clear walking path retained; no utility conflict	2, 3
2 Tree + bench	Shading; evapotranspiration	Resting	As street tree, plus seating outside the walking path	2, 3, 4
3 Tree in planter box	Shading	Walking, waiting	Hard surface where pits are impossible; irrigation	1
4 Climbing plant	Shading of façades; lower radiant heat	Walking	Façade or wall; small bed or container	1, 4
5 Vegetated pergola	Overhead shading	Waiting, resting	Space for structure; no deep soil needed	1, 4
Supporting tools				
6 Rain garden	Evapotranspiration (when moist); water retention	Supports trees	Unpaved strip or former parking bay; runoff inflow	2, 3
7 Permeable surface	Lower heat storage; infiltration	All	Low-traffic surfaces, parking bays, squares	All

Typologies: 1 enclosed tram streets, 2 wide open streets, 3 enclosed car-shared streets, 4 pedestrian streets and squares. Tools can be combined in other typologies where space allows. Source: adapted from World Bank (2021); spatial requirements and typologies by the author.

6.3 Positioning the Toolbox in the Turkish Context

As discussed in Section 2.9, nature-based solutions for heat in Türkiye have so far been implemented mainly at the scale of the city, the district or the large park (Table 6.3). The Istanbul Climate Change Action Plan relies on tree planting and new green areas without specifying where, in what form or for whom, the Millet Bahçeleri programme creates large new parks, and URBAN GreenUP in İzmir reported its effect on heat only qualitatively, while research in Istanbul has either modelled the city as a whole or tested single elements such as street trees.

Against this background, the toolbox contributes in three ways. It works within the existing street, drawing only on families that the catalogue rates suitable for small areas and high urban density, so that it does not depend on new land for parks. It selects tools through the processes that act at pedestrian level and the benefit that matters most for older adults, so that planting becomes a means of reducing the heat experienced while walking, waiting and resting rather than an end in itself. And it combines several families within a single street, so that where trees cannot be planted, climbing plants, pergolas and planter boxes take their place. Like most of these examples, however, its cooling effect has not been simulated, a limitation to which Chapter 8 returns.

Table 6.3. Nature-based solutions in Türkiye read through the catalogue categories, compared with the toolbox developed in this thesis. Source: author, based on the cited sources and World Bank (2021).

Example	Catalogue family	Scale	Main focus	Left unaddressed
Istanbul Climate Change Action Plan (İBB, 2021)	Urban Forests, Open Green Spaces, Green Corridors	City	Tree planting, green area expansion	Location, form and users of measures
Millet Bahçeleri (Şahin & İnan, 2024; Aktepe, 2025)	Open Green Spaces	City / district	Large new parks	Everyday streets; neighbourhood access
URBAN GreenUP, İzmir (CORDIS, 2021; URBAN GreenUP, 2025)	Green Corridors, Building Solutions, Bioretention Areas, River and Stream Renaturation	District / corridor	Multiple hazards, demonstration	Measured heat outcomes; dense historic fabric
Street trees in Istanbul canyons (Sarı et al., 2025)	Green Corridors	Street	Pedestrian thermal stress	Combination with other tools; older users
Stormwater NBS, Kadıköy (Abravanel, 2025)	Bioretention Areas	Neighbourhood	Runoff and flooding	Heat
This thesis: NBS toolbox	**Building Solutions, Open Green Spaces, Green Corridors + Bioretention Areas**	**Street**	**Heat experienced by older adults walking, waiting and resting**	**Cooling not simulated; tested on four sites**

Catalogue families as defined in World Bank (2021).
Source: Author, based on the cited sources and World Bank (2021).

7. Design

The design proposal applies the NBS toolbox to the Green Loop, translating the strategy into site-specific interventions across Caferağa and Osmanağa. Its aim is to develop a connected network of cooler everyday spaces and walking routes, with particular attention to the daily activities and mobility of older residents during hot summer conditions.

Following the selection in Section 5.4, four design sites are developed in detail: Sections A-A and C-C (Typology 1), Section E-E (Typology 3) and Section F-F (Typology 2), located along the Green Loop as shown in Figure 7.2, while the pedestrian streets of Typology 4 are addressed through the design guidance in Table 5.3. At each site, solar exposure and existing shade were considered alongside street dimensions, street function and planting capacity, to determine where additional vegetation, shading or changes to paved surfaces could be introduced, and which activities, whether walking, resting or waiting, the space needs to support.

Selected tools from the NBS toolbox are applied individually or in combination according to the conditions of each site, following the principles set out in Section 7.1, and the four designs are then compared in Section 7.6, which shows how the allocation of street width and the extent of sealed surface change with the interventions.

Section	Width	Walking zones before	Walking zones after	Planted and permeable before	Planted and permeable after	Vehicle and tram before	Vehicle and tram after
A-A Pedestrian–tram corridor	12.5 m	9.0 m (72%)	3.6 m (29%)	0	5.4 m (43%)	3.5 m (28%)	3.5 m (28%)
C-C Mixed tram street	10.0 m	4.0 m (40%)	3.6 m (36%)	0	1.0 m (10%)	6.0 m (60%)	5.4 m (54%)
E-E Parking-dominated street	13.5 m	4.0 m (30%)	4.8 m (36%)	0	5.4 m (40%)	9.5 m (70%)	3.3 m (24%)
F-F Boulevard street	19.2 m	4.0 m (21%)	6.6 m (34%)	1.2 m (6%)	6.6 m (34%)	14.0 m (73%)	6.0 m (31%)
All four sections	**55.2 m**	**21.0 m (38%)**	**18.6 m (34%)**	**1.2 m (2%)**	**18.4 m (33%)**	**33.0 m (60%)**	**18.2 m (33%)**

C-C: the 2.6 m pavement is divided into a 1.6 m walking zone and a 1.0 m zone of trees in planter boxes; the 2.0 m tram-stop pavement is shaded by a vegetated pergola.
Planted and permeable zones: planted strips, rain gardens, planter and bench zones and the median. Vehicle space includes parking and the tram track.
Source: Author, measured from the design sections.

Figure 7.1. Vision of a shaded tram street on the Green Loop. Source: author.

7.1 Design Principles

The interventions influence heat through three complementary mechanisms. Tree canopies and vegetated pergolas intercept direct solar radiation before it reaches pedestrians and the ground, which is the most significant factor in reducing thermal stress during the day, since mean radiant temperature dominates outdoor comfort in hot conditions (Section 2.4). Vegetation also cools the surrounding air through evapotranspiration, although this effect is limited in narrow streets and during dry periods. Replacing asphalt and concrete with planted and permeable surfaces reduces the heat stored in the street during the day and released at night, while rain gardens retain water in the soil that sustains planting through the summer. Because shade has the most direct effect on the conditions older adults experience while walking, waiting and resting, it is treated as the primary objective at each site, with surface modification as a supporting measure.

The tools are assigned to each site through six principles that translate the street criteria of Chapter 5 into design decisions, so that the response at each site follows from its conditions rather than from a standard solution.

Shade where people are most exposed. Shade is placed first where pedestrian space receives the most direct sunlight and where older adults wait or rest, such as tram stops, benches and shop frontages.

Trees where the ground allows, alternatives where it does not. Trees are planted in the ground where pavement width and underground conditions allow sufficient root volume; where tram infrastructure, utilities or basements prevent this, they are planted in planter boxes, and where no tree fits, climbing plants and vegetated pergolas provide shade.

Canopy scaled to enclosure. In open streets, large-canopy trees compensate for the lack of building shade, while in deeper canyons smaller trees and vertical greening are used so that ventilation is not obstructed.

Street space reallocated from parking. Where parking occupies the street edges, parking lanes become planted strips, rain gardens and seating, while access for deliveries and emergency vehicles is maintained.

Supporting tools placed with primary ones. Rain gardens and permeable surfaces are placed beside and over the root zones of trees, so that the water they retain sustains the planting that provides shade; where they adjoin walking zones, smooth, narrow-jointed pavers avoid tripping hazards for older adults with walking aids.

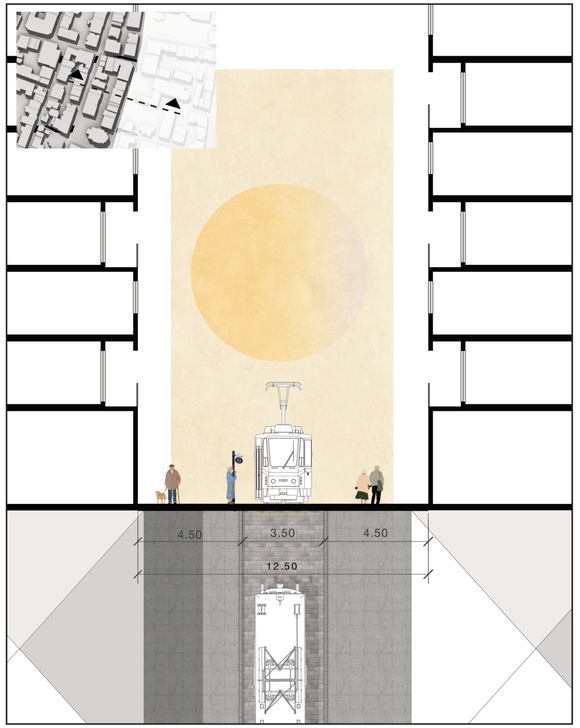
Comfort throughout the year. Deciduous species are preferred, so that streets are shaded in summer while winter sunlight reaches pavements and façades, and broad-canopied species such as Platanus orientalis are used where space allows, given their cooling effect in Istanbul's street canyons (Sarı et al., 2025).

Figure 7.2. The Green Loop in Caferağa and Osmanağa, with the ten surveyed sections and the four design sites (A-A, C-C, E-E, F-F). Source: author, based on OpenStreetMap data.

Figure 4.13b. Locations of the thirteen morphology samples shown in Figure 4.14. Source: author's analysis in Google Earth Engine and QGIS; data: USGS Landsat 8 and 9 Collection 2, Level-2.

7.2 Section A-A: Pedestrian–Tram Corridor

Section A-A represents the enclosed tram streets of Typology 1. The 12.5 m street is lined by buildings of about 21 m (H/W 1.7) and runs north–south, so that its pedestrian space is exposed to direct sunlight around the middle of the day, when the buildings cast little shade. Before intervention, two 4.5 m pavements flanked the 3.5 m tram track, with only scattered trees. The design divides each pavement into a 1.8 m walking zone along the façades and a 2.7 m planted zone beside the tramway. Because the tram's underground infrastructure limits in-ground planting, trees are placed in planter boxes, while a single in-ground tree with a bench creates a resting point beside the tram. Climbing plants on both façades reduce the heat radiated towards the walking zones, and the planted zones are surfaced with permeable paving, forming a continuous band of shade on both sides of the tram.



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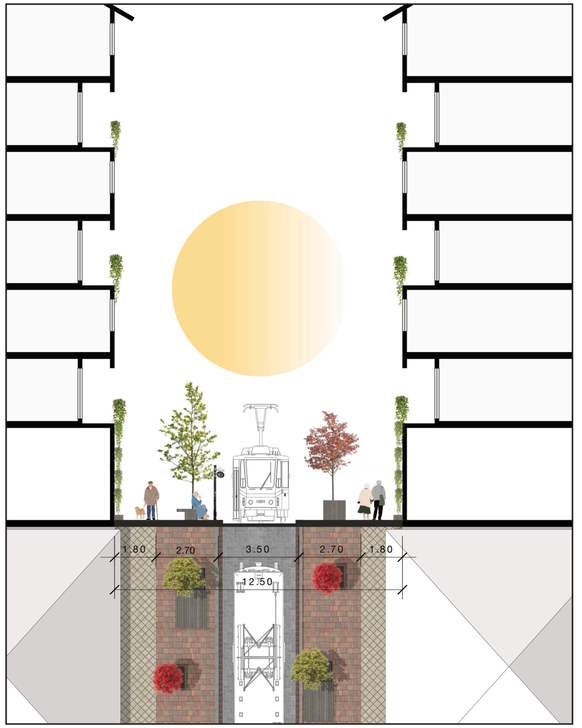
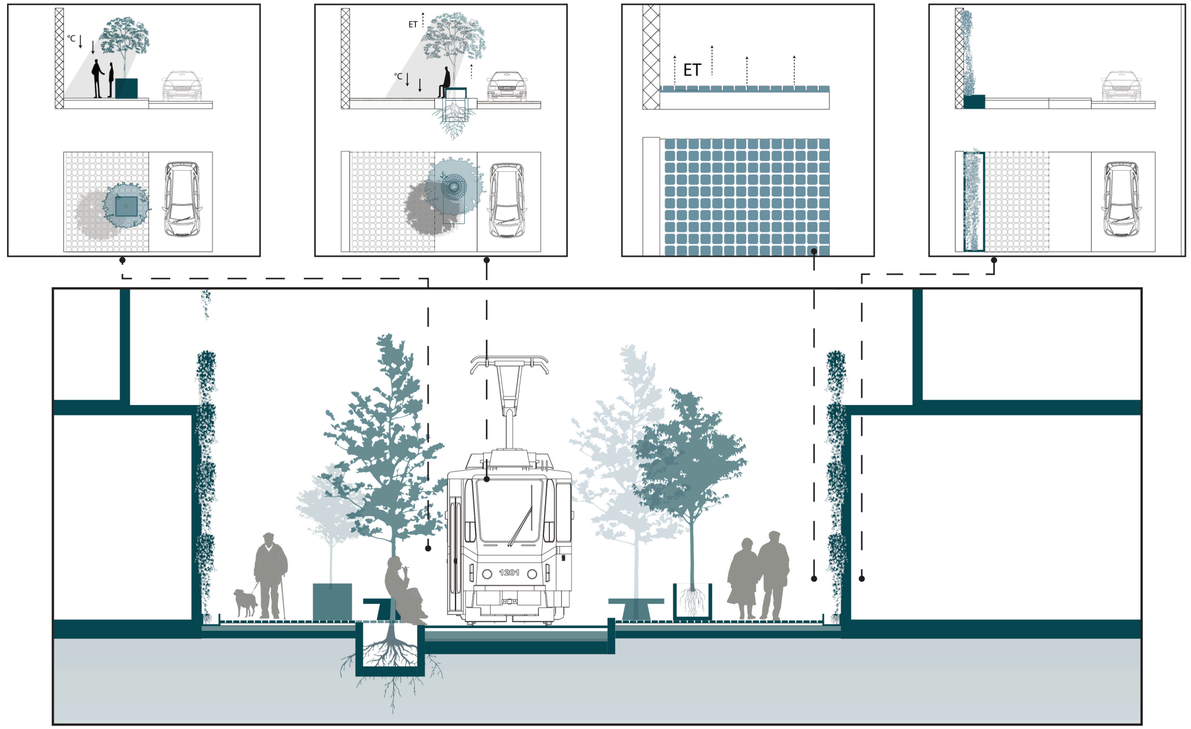


After

Detail Section



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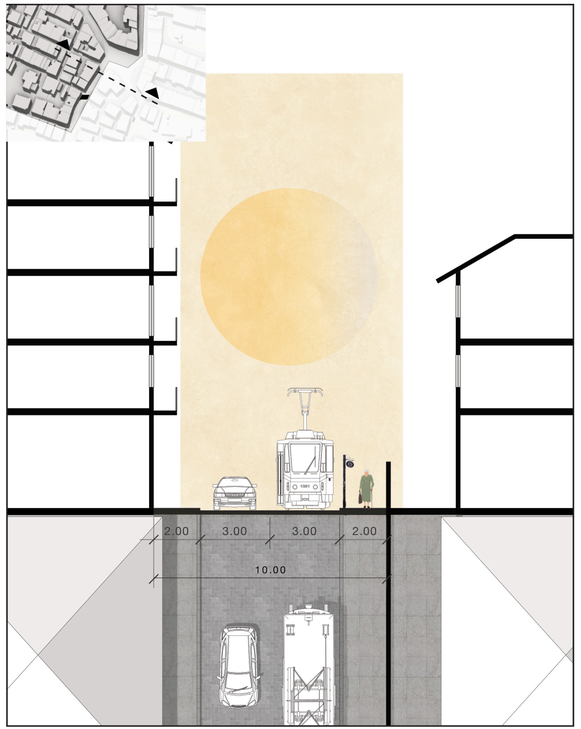
7.3 Section C-C: Mixed Tram Street

Section C-C is the second tram street of Typology 1 and shows how the response developed for A-A is adjusted to a narrower street. The 10 m street is lined by buildings of about 15 m (H/W 1.5) and runs north-east to south-west; before intervention, a 3 m traffic lane and the 3 m tram track occupied 6 m of its width, leaving two 2 m pavements, one of which serves as a tram stop, with only scattered street trees, and its pedestrian space was exposed to direct sunlight for much of the day on 23 July. As there is no room for a planted zone on both sides, the traffic lane is narrowed to 2.4 m, which allows the opposite pavement to be widened to 2.6 m, where trees in planter boxes line a 1.6 m walking zone and climbing plants green the façade. At the tram stop, where older adults wait, a vegetated pergola shades the 2 m platform, so that shade is concentrated where it is needed most rather than spread thinly across a street too narrow for a continuous canopy.



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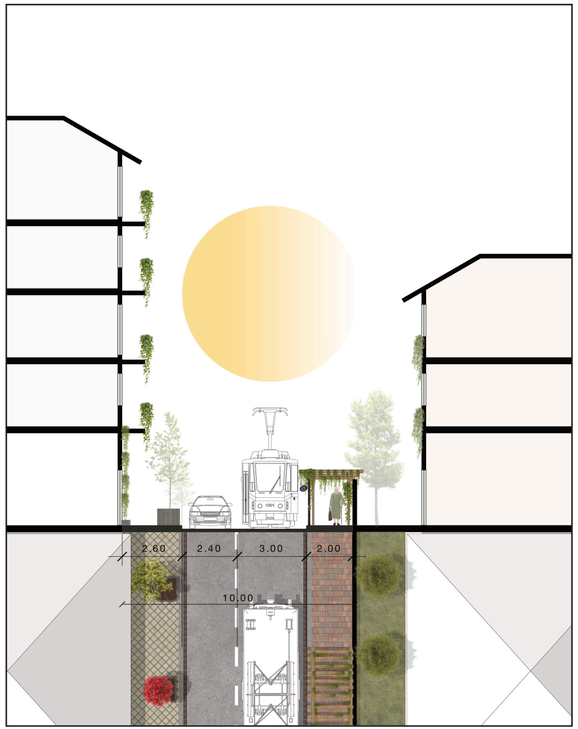

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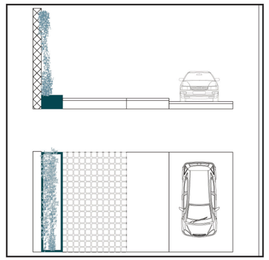
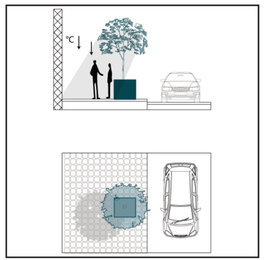
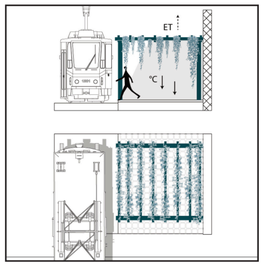
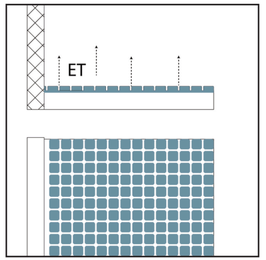
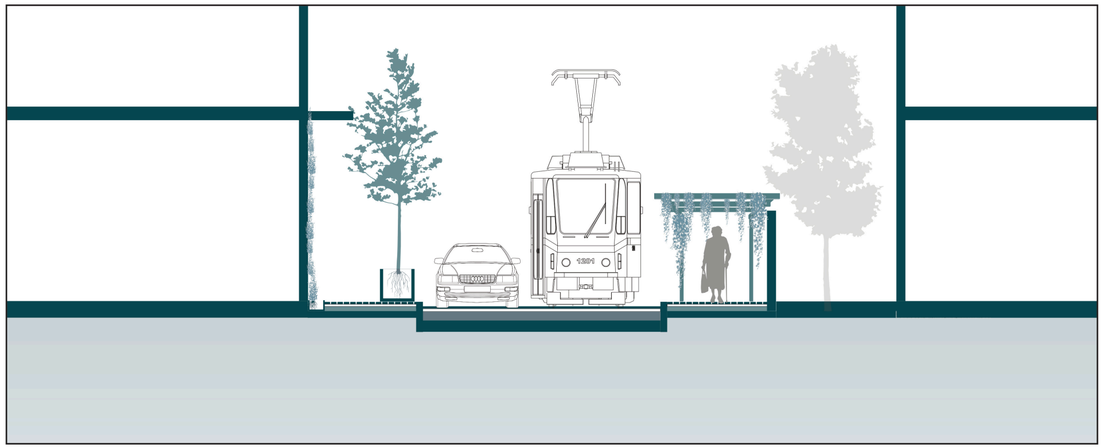




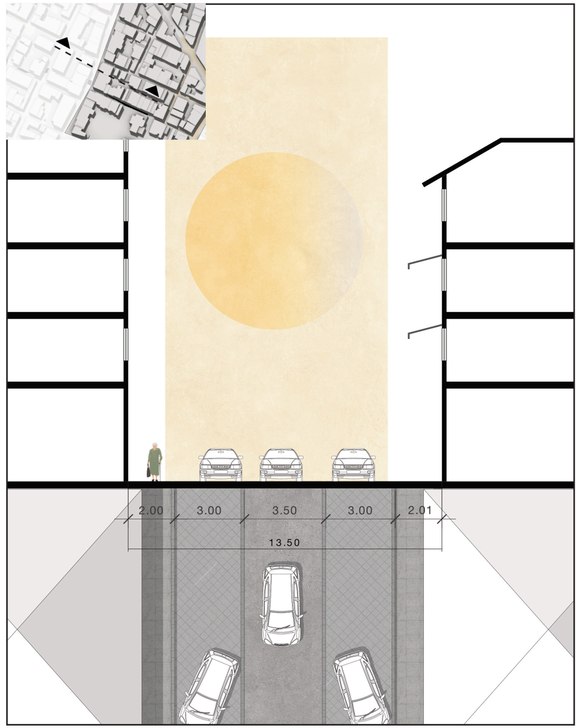






7.4 Section E-E: Parking-Dominated Street

Section E-E represents the car-shared streets of Typology 3. The 13.5 m street is lined by buildings of about 18 m (H/W 1.3) and runs north-north-east to south-south-west; before intervention, parking on both sides and a 3.5 m carriageway occupied 9.5 m of its width, leaving 2 m pavements with no vegetation and almost entirely sealed ground, and the street was highly exposed to direct sunlight on 23 July. The design removes on-street parking, converts the two parking lanes into 2.7 m planted strips, widens the pavements to 2.4 m and retains a 3.3 m carriageway for access. The planted strips hold trees, in the ground where possible and otherwise in planter boxes, with benches in their shade and permeable surfaces around them, while climbing plants on the façades add shade at pedestrian height where the street is too narrow for large canopies. In this typology, parking space is the main reserve of land for cooling.



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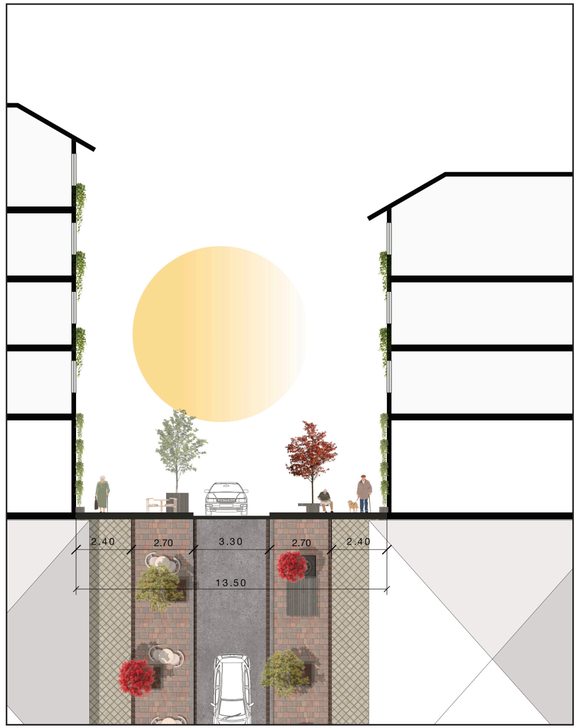

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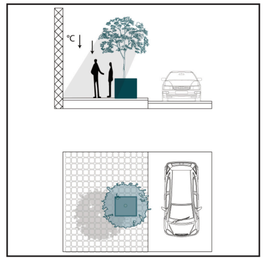
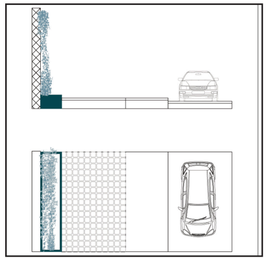
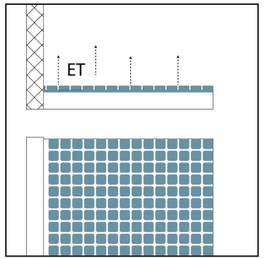
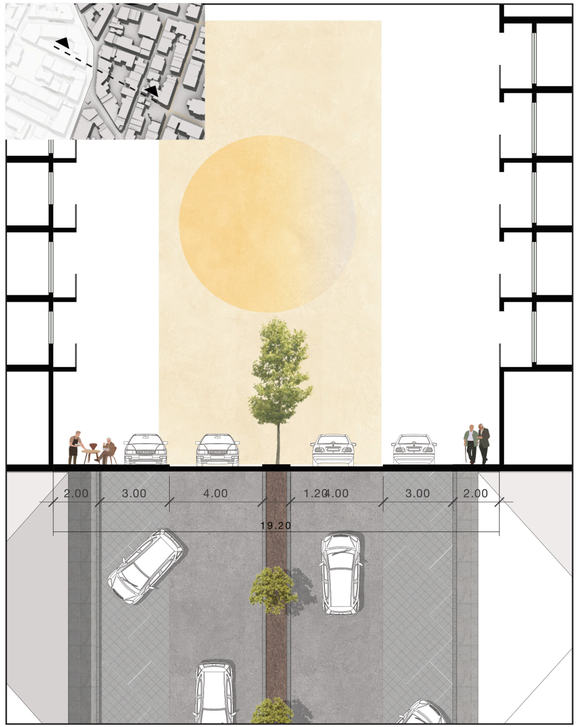






7.5 Section F-F: Boulevard Street

Section F-F represents the wide open streets of Typology 2, where buildings provide little shade and comfort depends on the continuity of the tree canopy. The 19.2 m street (H/W 0.9) runs north–south; before intervention, two parking lanes and two 4 m carriageways occupied 14 m of its width, leaving 2 m pavements, and shade came only from a single row of trees in the 1.2 m median. The design removes on-street parking, narrows the carriageway to one 3 m lane in each direction and widens the pavements to 3.3 m. The former parking lanes become 2.7 m planted strips with rain gardens and trees, which collect runoff from the road and keep the soil moist, while the median trees are retained. Benches beneath trees in planter boxes provide resting points and climbing plants green the façades, so that three parallel lines of canopy now run along a street that was previously exposed for most of the day.



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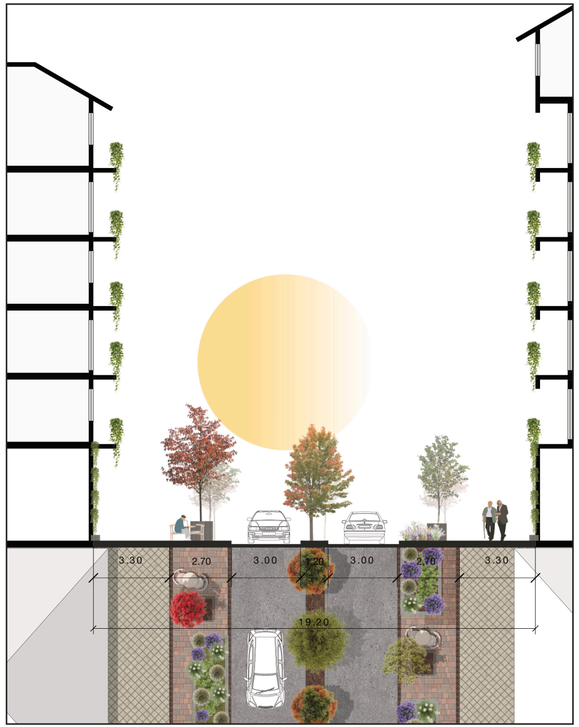

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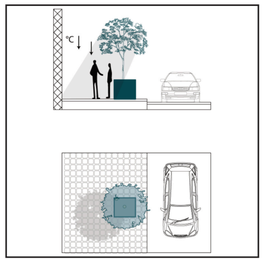

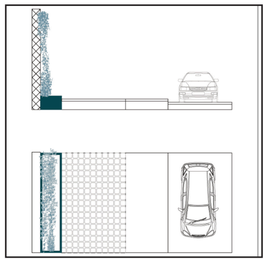

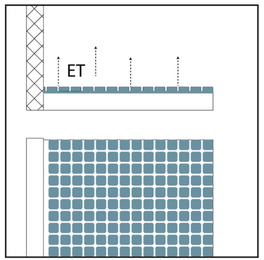

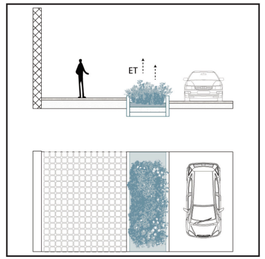

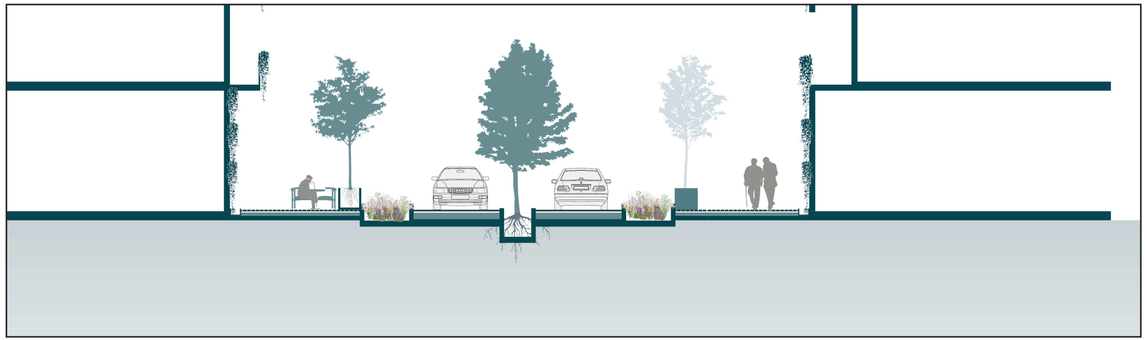

8. Discussion and Reflection

8.1 Discussion

The development of this framework reinforced the importance of considering both social and physical context when integrating nature-based solutions into existing urban environments. General cooling principles such as shade, vegetation, evapotranspiration and the modification of impervious surfaces provide a starting point, but they do not determine where intervention should be prioritised or how individual measures should respond to local conditions. Throughout the research, these two questions became increasingly connected: the social context identified where cooling could support everyday life, while the physical and environmental context informed how NBS could be integrated.

The findings are largely consistent with the literature reviewed in Chapter 2. As Ali-Toudert and Mayer (2006) and Johansson (2006) showed, deep and narrow streets can provide considerable self-shading, and the solar analysis in this thesis likewise found that the most enclosed sections, D-D and J-J, received the least direct sun, while the most exposed included the open square, the boulevard and the streets whose edges are occupied by parking, where, as Coutts et al. (2016) found, additional tree shade brings the greatest benefit. The analysis also showed that enclosure alone does not predict exposure: Section A-A, although enclosed (H/W 1.7), was among the most exposed because its north–south orientation admits the high midday sun, whereas D-D, an east–west street, remained shaded largely because it is very narrow (H/W 2.3). This qualifies the finding that east–west streets in Istanbul experience the highest thermal stress (Sarı et al., 2025) and confirms that orientation and enclosure must be read together. Finally, the framework extends the prioritisation approach of Norton et al. (2015): rather than identifying priority neighbourhoods by where vulnerable people live, it identifies priority streets by where they walk, wait and rest, a shift that the Barcelona experience suggests is necessary, since proximity to cool places does not guarantee that older adults can reach them (Meza et al., 2026).

An important outcome of the design process was the recognition that the Green Loop does not require continuous physical intervention, because existing environmental qualities matter as much as new measures. The surroundings of the Caferağa Mosque, for example, already benefit from established vegetation and shade and were therefore given lower priority than more exposed sections, which highlighted the value of preserving existing cooling assets rather than treating every part of the network as a site for new intervention. At the same time, some relevant places were not developed as design sites: the pedestrian streets and squares of Typology 4, including the shared market street H-H, and destinations such as the Balık Pazarı market are addressed only through design guidance. This reflects the limits of the project rather than an absence of potential, and the toolbox could be applied to them by considering their activities, spatial configuration, solar exposure, existing vegetation and available space.

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The four design sites therefore demonstrate selected ways in which NBS can respond to different conditions, but not the full extent of possible applications. Similar functions and spatial conditions, including neighbourhood markets, residential and commercial streets, public transport stops and places of worship, occur elsewhere in Kadıköy and in other neighbourhoods, and the transferable aspect of the framework is the process of identifying where cooling is needed and adapting interventions to local conditions, rather than the reproduction of the proposed designs.

8.2 Limitations

The research has several limitations. It addresses only part of the wider urban heat challenge, since anthropogenic heat from traffic remains relevant in Caferağa and Osmanağa and a comprehensive restructuring of traffic circulation falls outside its scope. Its environmental methods describe surface temperature and direct sun on a single representative day rather than the thermal comfort of pedestrians across the summer, and the cooling performance of the designs was not simulated, so that field measurements and microclimate modelling using indices such as PET or UTCI are needed to quantify their effect. The everyday destinations of older adults were identified through an exploratory interview rather than with older residents themselves, and could be verified through participatory mapping and accompanied walks. Finally, the designs do not assess the maintenance and irrigation demand of the proposed planting, on which the long-term viability of the toolbox depends during increasingly dry summers.

9. Conclusions

This thesis set out to answer how nature-based solutions can be contextualised to mitigate urban heat while supporting the everyday mobility of an ageing population. Starting from the observation that NBS for heat are often reduced in practice to generic tree planting, it developed a framework in which the social context informs where cooling is prioritised and the physical context informs how it is delivered, and tested this framework in Caferağa and Osmanağa, the historic centre of Kadıköy, Istanbul.

9.1 Answering the Research Questions

Where should urban cooling be prioritised? The district-scale analysis showed that surface heat in Kadıköy is associated less with particular land uses than with recurring urban morphologies, and that the most intense hotspots, on transport corridors, port land and large single-use sites, lie outside the spaces that older adults use every day. Cooling for older adults should therefore be prioritised not where surface heat is highest, but where elevated heat coincides with the destinations and routes of everyday life. In the study area these destinations are concentrated along Bahariye, Moda and Söğütlüçeşme Caddesi, and the Green Loop that connects them passes within 100 m of around two-thirds of the mapped destinations, so that cooling this network serves the largest number of daily trips.

How do local conditions influence the selection and adaptation of NBS? Enclosure, orientation, street function and planting capacity varied considerably along the loop, and together they determined both whether intervention was needed and which measures were feasible. The narrowest sections were already shaded by their buildings, while the open square, the boulevard and the streets lined with parking were the most exposed. Grouping the ten surveyed sections into four typologies made these differences operational: in enclosed tram streets, tram infrastructure rules out most in-ground planting and shade comes from planter boxes and climbing plants; in wide open streets, comfort depends on completing the tree canopy; in car-shared streets, parking space is the main reserve for planting; and in pedestrian streets, narrow widths and market activity call for overhead and vertical greening. The catalogue analysis showed that only four of the World Bank's fourteen NBS families cool through processes that act at pedestrian level and fit within dense fabric, which explains why a context-specific toolbox, rather than a general catalogue, is needed.

How can context-specific NBS be integrated into a connected cooling network? The four actions of the strategy, Anchor, Cool, Connect and Expand, establish an order of priority: shade first where older adults wait and rest, then along the routes between destinations, and finally in the residential streets that lead into them. Within this sequence, each typology provides a design response that can be transferred from its design site to the other sections of the same typology and, later, to the side streets. The network is thus built from repeated, condition-specific responses rather than from a single uniform intervention, and across the four design sections, the share of street width given to planting and permeable surfaces rises from 2% to 35%.

The main research question. NBS can be contextualised to mitigate urban heat while supporting the everyday mobility of older adults by deciding where to intervene according to how older adults use the city, and how to intervene according to the conditions of each street. Shade is the primary objective, because mean radiant temperature dominates daytime thermal stress for pedestrians, while evapotranspiration and permeable surfaces play a supporting role. Where trees can be planted they provide this shade, and where they cannot, climbing plants, pergolas and planter boxes take their place, so that cooling is not abandoned in the narrow and constrained streets where older adults often walk

9.2 Transferability

The framework is transferable as a process rather than as a set of designs. Other districts of Istanbul, and other dense Mediterranean cities with ageing populations and historic street networks, could follow the same sequence: screen surface heat with freely available satellite data; distinguish hotspots in the fabric where people live and walk from those on large single-use sites; map the everyday destinations of the group in question; connect them into a priority network; classify its streets into typologies; and select NBS according to their cooling processes and their suitability for the local density. The social lens can also be changed: for children, the destinations would be schools, playgrounds and the routes between them; for outdoor workers, transport stops and workplaces; and for people with disabilities, accessible routes and resting points. In each case, where people are exposed during everyday life, and not only where they live, defines where cooling should begin.

9.3 Contribution and Future Research

The main contribution of the thesis lies not in a fixed collection of interventions but in an approach for deciding where and how NBS can be integrated into an existing neighbourhood. It shifts the unit of prioritisation from the neighbourhood to the everyday network of streets and destinations, and it links the selection of NBS to the measurable conditions of each street, offering an alternative to the generic tree planting that dominates current policy in Istanbul. Its limitations point to the next steps: the cooling performance of the designs should be tested through microclimate simulation or field measurement, using indices such as PET or UTCI, and older adults themselves should be involved in identifying their routes, resting places and experiences of heat.

As heatwaves become more frequent and populations continue to age, cities will need to cool not only their hottest places but the places where the most vulnerable people spend their days. This thesis suggests that doing so requires nature-based solutions chosen for the street in which they are planted and for the people who walk along it.

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